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5 **Draft IEEE Standard for**
6 **Local and metropolitan area networks—**

7 **Timing and Synchronization for**
8 **Time-Sensitive Applications**

9 **Amendment: Support for the IEEE Std 802.3**
10 **Clause 4 Media Access Control (MAC) operating in**
11 **half-duplex**

12 Prepared by the
13 **Time-Sensitive Networking (TSN) Task Group of IEEE 802.1**

14 Sponsor
15 **LAN/MAN Standards Committee**
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17 **IEEE Computer Society**

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1 **Abstract:** This amendment to IEEE Std 802.1AS™-2020 specifies protocols, procedures, and
2 managed objects that support IEEE Std 802.3 Clause 4 Media Access Control (MAC) operating in
3 half-duplex while retaining existing functionality and backward compatibility, and remaining a profile
4 of IEEE Std 1588™-2019.

5 This amendment addresses errors and omissions in the description of existing functionality.

6 **Keywords:** best timeTransmitter, frequency offset, Grandmaster Clock, Grandmaster PTP
7 Instance, PTP End Instance, PTP Relay Instance, IEEE Std 802.1AS™, phase offset,
8 synchronization, syntonization, time-aware system

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Silvana Rodrigues, *Editor P802.1ASds*

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1 Introduction

This introduction is not part of IEEE Std 802.1ASdsTM-20xx, IEEE Standard for Local and metropolitan area networks—Timing and Synchronization for Time-Sensitive Applications—Amendment: Support for the IEEE Std 802.3 Clause 4 Media Access Control (MAC) operating in half-duplex

2 The first edition of IEEE Std 802.1AS was published in 2011. A first corrigendum, IEEE Std
3 802.1ASTM-2011/Cor1-2013, provided technical and editorial corrections. A second corrigendum, IEEE Std
4 802.1ASTM-2011/Cor2-2015 provided additional technical and editorial corrections.

5 The second edition, IEEE Std 802.1AS-2020, added support for multiple gPTP domains, Common Mean
6 Link Delay Service, external port configuration, and Fine Timing Measurement for 802.11 transport.
7 Backward compatibility with IEEE Std 802.1AS-2011 was maintained. A corrigendum, IEEE Std
8 802.1ASTM-2020/Cor1-2021, provides technical and editorial corrections.

9 The third edition, IEEE Std 802.1AS-2025 is a roll-up of IEEE Std 802.1AS-2020 with the corrigendum
10 IEEE Std 802.1AS-2020/Cor1, and its amendments: IEEE Std 802.1ASdr, IEEE Std 802.1ASdn, and IEEE
11 Std 802.1ASdm.

12 This amendment to IEEE Std 802.1AS-2025 specifies protocols, procedures, and managed objects that
13 support IEEE Std 802.3 Clause 4 Media Access Control (MAC) operating in half-duplex while retaining
14 existing functionality and backward compatibility, and remaining a profile of IEEE Std 1588TM-2019.

15 This amendment addresses errors and omissions in the description of existing functionality

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2 **Draft IEEE Standard for** **Local and metropolitan area networks—**

4 **Timing and Synchronization for Time-** 5 **Sensitive Applications**

6 **Amendment: Support for the IEEE Std 802.3** **Clause 4 Media Access Control (MAC) operating in** **half-duplex**

9 (This amendment is based on IEEE Std 802.1AS™-2025.)

10 NOTE—The editing instructions contained in this amendment define how to merge the material contained therein into
 11 the existing base standard and its amendments to form the comprehensive standard.

12 The editing instructions are shown in ***bold italic***. Four editing instructions are used: change, delete, insert, and replace.
 13 ***Change*** is used to make corrections in existing text or tables. The editing instruction specifies the location of the change
 14 and describes what is being changed by using ~~striketrough~~ (to remove old material) and underscore (to add new
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 16 and insertions may require renumbering. If so, renumbering instructions are given in the editing instruction. ***Replace*** is
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 18 one. Editing instructions, change markings, and this NOTE will not be carried over into future editions because the
 19 changes will be incorporated into the base standard.¹

20

¹Notes in text, tables, and figures are given for information only, and do not contain requirements needed to implement the standard.

3. Definitions

Insert the following definitions in Clause 3, and renumber the definitions as appropriate:

3.17 Half-duplex Ethernet: IEEE Std 802.3TM Clause 4 Media Access Control (MAC) operating in half-duplex.

1 4. Acronyms and abbreviations

2 *Insert the following acronym in clause 4 as follows:*

3 HDE Half-duplex Ethernet

5. Conformance

Change the title of 5.5 as follows:

5.5 .MAC-specific timing and synchronization methods for full-duplex point-to-point IEEE 802.3 links

Change the first sentence of 5.5 as follows:

An implementation of a time-aware system with full-duplex IEEE Std 802.3-2022 media access control (MAC) services to physical ports shall:

Change the sentence after item e) of 5.5 as follows:

An implementation of a PTP Instance with full-duplex IEEE Std 802.3-2022 MAC services to physical ports may:

Insert 5.9 as follows:

5.9 MAC-specific timing and synchronization methods for half-duplex IEEE 802.3 (HDE) links

An implementation of a time-aware system with half-duplex IEEE Std 802.3-2022 media access control (MAC) services to physical ports shall:

- a) Support half-duplex operation, as specified in Clause 4 of IEEE Std 802.3-2022.
- b) Support the requirements as specified in Clause 19.

18

17. Time-synchronization model for a packet network

2 7.2 Architecture of a time-aware network

3 7.2.1 General

4 *Add item g) after item f) in the lettered list in 7.2.1 as follows:*

- 5 g) IEEE Std 802.3 Clause 4 Media Access Control (MAC) operating in half-duplex as specified in
6 Clause 19.

7 7.6 Differences between gPTP (IEEE Std 802.1AS) and PTP (IEEE Std 1588-2019)

8 *Change item e) and item f) in 7.6 as follows:*

- 9 e) For full-duplex Ethernet links and HDE links, gPTP requires the use of the peer-to-peer delay
10 mechanism, while IEEE Std 1588-2019 also allows the use of end-to-end delay measurement.
- 11 f) For full-duplex Ethernet links and HDE links, gPTP requires the use of two-step processing (use of
12 Follow_Up and Pdelay_Resp_Follow_Up messages to communicate timestamps), with an optional
13 one-step processing mode for full-duplex Ethernet that embeds timestamps in the Sync “on the fly”
14 as they are being transmitted (gPTP does not specify one-step processing for peer delay messages).
15 IEEE Std 1588-2019 allows either two-step or one-step processing to be required (for both Sync and
16 peer delay messages) depending on a specific profile.

1 8. IEEE 802.1AS concepts and terminology

2 8.5 Ports

3 8.5.1 General

4 *Change 8.5.1 as follows:*

5 The PTP Instances in a gPTP domain interface with the network media via physical ports. gPTP defines a
6 ~~logical port, i.e.,~~ a PTP Port, in such a way that communication between PTP Instances is point-to-point or,
7 point-to-multipoint. The choice of point-to-point (e.g., see Clause 16) or point-to-multipoint (e.g., see
8 Clause 19) communication is specified in each media dependent specification in this standard. even over
9 ~~physical ports that are attached to shared media. One logical~~ A PTP port, consisting of one PortSync entity
10 and one media-dependent (MD) entity, ~~is instantiated for each PTP Instance with which the PTP Instance~~
11 ~~communicates. For shared media, multiple logical ports can be associated with a single physical port.~~

12 ~~Unless otherwise qualified, each instance of the term port refers to a logical port.~~

1 10. Media-independent layer specification

2 10.2 Time-synchronization state machines

3 10.2.4 Per PTP Instance global variables

4 10.2.4.28 driftTrackingTlvSupport

5 *Change the note in 10.2.4.28 as follows:*

6 NOTE—The Drift_Tracking TLV is transported only on full-duplex, point-to-point links (as specified in Clause 11), and
7 HDE links (as specified in Clause 19). Even if driftTrackingTlvSupport is TRUE, the Drift_Tracking TLV is not
8 transported on links other than full-duplex, point-to-point links, and HDE links, and is not received by the PTP Ports at
9 the other end of these links.

10 10.2.5 Per-port global variables

11 10.2.5.1 asCapable

12 *Change the note in 10.2.5.1 as follows:*

13 NOTE—For full-duplex point-to-point links, ~~The per-port global variable~~ asCapableAcrossDomains (see 11.2.13.13) is
14 common across, and accessible by, all the domains. For HDE links, asCapableAcrossDomains is a per-PTP Instance
15 (i.e., per domain). ~~asCapableAcrossDomains~~ is computed by the MDPdelayReq state machine (see 11.2.19). For full-
16 duplex point-to-point links (see Clause 11) and HDE links (see Clause 19), asCapableAcrossDomains is used when
17 setting the instance of asCapable for each domain (for the link in question).

18 10.5 Message attributes

19 10.5.3 Addresses

20 *Change the last paragraph in 10.5.3 as follows:*

21 If the transport is full-duplex or half-duplex IEEE 802.3, all Announce and Signaling messages transmitted
22 shall use the MAC address of the respective egress physical port as the source address.

23 10.6 Message formats

24 10.6.4 Signaling message

25 10.6.4.3 Message interval request TLV definition

26 10.6.4.3.9 flags (Octet)

27 *Change the note in 10.6.4.3.9 as follows:*

28 NOTE—For full-duplex point-to-point links (see Clause 11) and HDE links (see Clause 19), it is expected that the PTP
29 Port sending this TLV will set bits 0 and/or 1 to FALSE if this PTP Port will not provide valid timing information in its
30 subsequent responses (Pdelay_Resp and Pdelay_Resp_Follow_Up) to Pdelay_Req messages. Similarly, it is expected
31 that the PTP Port sending this TLV will set bit 2 to TRUE if it is capable of receiving and correctly processing one-step
32 Sync messages.

11. Media-dependent layer specification for full-duplex point-to-point links

11.1 Overview

11.1.1 General

Add a paragraph at the end of 11.1.1 as follows:

PTP links using the IEEE Std 802.3 Clause 4 MAC operating in half-duplex mode are specified in Clause 19. Since clause 19 references some of the state machines defined in this clause, Clause 11 also includes the state machine variables and conditions relevant to links operating in half-duplex mode

11.2 State machines for MD entity specific to full-duplex point-to-point links

11.2.1 General

Replace Figure 11-5 with the following:

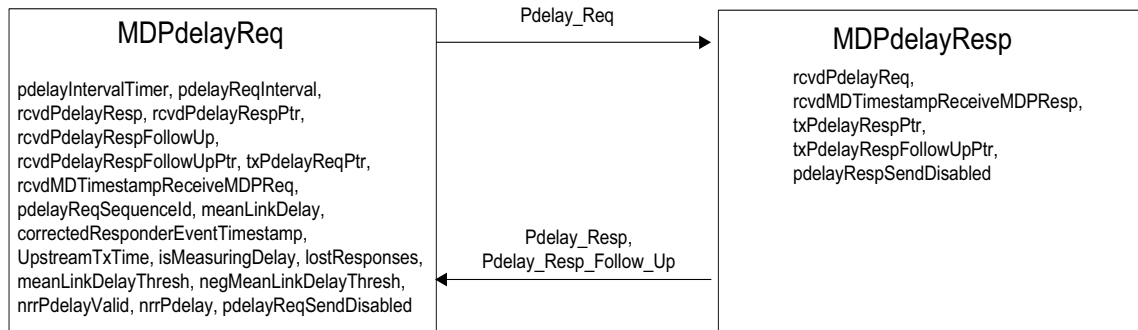


Figure 11-5—Peer-to-peer delay mechanism state machines—overview and interrelationships

11.2.2 Determination of asCapable and asCapableAcrossDomains

Change 11.2.2 as follows:

There is one instance of the global variable asCapable (see 10.2.5.1) per PTP Port, per domain. For full-duplex point-to-point links, there is one instance of the global variable asCapableAcrossDomains (see 11.2.13.13), per port, that is common across, and accessible by, all the domains. For HDE links, there is one instance of asCapableAcrossDomains per PTP Instance (i.e., per domain) (see 11.2.13.13).

The ~~per PTP Port~~ global variable asCapable (see 10.2.5.1) indicates whether the IEEE 802.1AS protocol is operating, in this domain, on the PTP Link attached to this PTP Port, and can provide the required time-synchronization performance ~~described in B.3~~. asCapable is used by the PortSync entity, which is media-independent; however, the determination of asCapable is media-dependent.

The per-port global variable asCapableAcrossDomains is set by the MDPdelayReq state machine (see 11.2.19 and Figure 11-9). For a port attached to a full-duplex point-to-point PTP Link or to an HDE link, asCapableAcrossDomains shall be set to TRUE if and only if either:

- 1 a) ~~It~~ is determined, via the transport-specific peer-to-peer delay mechanism or CMLDS, that the
 2 following conditions hold for the port:
- 3 1) ~~+~~The port is exchanging peer delay messages with its neighbor,
 4 2) ~~b)~~The measured delay does not exceed meanLinkDelayThresh__or is below
 5 negMeanLinkDelayThresh,
 6 3) ~~+~~For full-duplex links, ~~+~~the port does not receive multiple Pdelay_Resp or
 7 Pdelay_Resp_Follow_Up messages in response to a single Pdelay_Req message, ~~and~~
 8 4) For HDE links, the port does not receive multiple Pdelay_Resp or multiple
 9 Pdelay_Resp_Follow_Up messages with the requestingPortIdentity.Clock Identity equal to the
 10 PTP Instance's clock identity in response to a single Pdelay_Req message, and
 11 5) ~~+~~The port does not receive a response from itself or another PTP Port of the same PTP
 12 Instance.
- 13 or:
- 14 b) pdelayReqSendDisabled is set to TRUE.

15 NOTE 1—If a PTP Instance implements only domain 0 and the MDPdelayReq and MDPdelayResp state machines are
 16 invoked on domain 0 (see 11.2.19), asCapableAcrossDomains is still set by the MDPdelayReq state machine.

17 The default value of meanLinkDelayThresh shall be set as specified in Table 11-1.

Table 11-1—Value of meanLinkDelayThresh for various links

Link	Value of meanLinkDelayThresh (ns) (see NOTE)
100BASE-TX, 1000BASE-T	800 ₁₀
100BASE-FX, 1000BASE-X	0x7FFFFFFF
NOTE—The actual propagation delay for 100BASE-TX and 1000BASE-T links is expected to be smaller than the above respective threshold. If the measured mean propagation delay (i.e., meanLinkDelay; see 10.2.5.8) exceeds this threshold, it is assumed that this is due to the presence of equipment that does not implement gPTP. For 100BASE-FX and 1000BASE-Xlinks, the actual propagation delay can be on the order of, or larger than, the delay produced by equipment that does not implement gPTP; therefore, such equipment cannot be detected by comparing measured propagation delay with a threshold. In this case, meanLinkDelayThresh is set to the largest possible positive value (i.e., in binary notation a zero followed by all 1s). The default value for negMeanLinkDelayThresh is not defined and it is implementation specific. <u>This table does not list propagation delay threshold values for all types of PHYs because the values are implementation specific and depend on the application.</u>	

18 The per-PTP Port, per-domain global variable asCapable shall be set to TRUE if and only if the following
 19 conditions hold:

- 20 c) ~~+~~The value of asCapableAcrossDomains is TRUE, and
 21 d) ~~+~~One of the following conditions holds:
- 22 1) The value of neighborGptpCapable for this PTP Port is TRUE, or
 23 2) The value of domainNumber is zero, and the value of sdoId for peer delay messages received
 24 on this PTP Port is 0x100.

25 NOTE 2—Condition ~~+~~d) 2) ensures backward compatibility with the 2011 edition of this standard. A PTP Instance
 26 compliant with the current edition of this standard that is attached, via a full-duplex point-to-point PTP Link, to a PTP

1 Instance compliant with the 2011 edition of this standard will not receive Signaling messages that contain the gPTP
2 capable TLV and will not set neighborGtpCapable to TRUE. However, condition ~~4~~ d) 2) ensures that asCapable for
3 this PTP Port and domain (i.e., domain 0) will still be set to TRUE if condition c) holds because the peer delay messages
4 received from the time-aware system compliant with the 2011 edition of this standard will have sdold set to 0x100.

5 11.2.12 Overview of MD entity global variables

6 *Add the following note in Table 11-2:*

7 NOTE — asCapableAcrossDomains is per PTP Instance for HDE links.

8 11.2.13 MD entity global variables

9 *Change 11.2.13.13 as follows:*

10 **11.2.13.13 asCapableAcrossDomains:** A Boolean that is TRUE if and only if either: 11.2.2 item a) or
11 11.2.2 item b) is satisfied~~conditions a) through d) of 11.2.2 are satisfied. This Boolean~~
12 asCapableAcrossDomains is set by the MDPdelayReq state machine and is used in determining asCapable
13 for a port (see 11.2.2). There is one instance of this variable for all the domains (per port) for full-duplex
14 point-to-point links.~~And~~ the variable is accessible by all the domains. There is one instance of this variable
15 per PTP Instance (i.e., per domain) for HDE links (see 19.2.2). When only one domain is active,
16 asCapableAcrossDomains is equivalent to the variable asCapable (see 10.2.5.1).

17 11.2.19 MDPdelayReq state machine

18 11.2.19.2 State machine variables

19 *Change 11.2.19.2.2 as follows:*

20 11.2.19.2.2 rcvdPdelayResp:

- 21 a) For full-duplex point-to-point mode: A Boolean variable that notifies the current state machine
22 when a Pdelay_Resp message is received. This variable is reset by the current state machine.
- 23 b) For half-duplex mode: A Boolean variable that notifies the current state machine when a
24 Pdelay_Resp message is received AND its requestingPortIdentity.clockIdentity is equal to the
25 clockIdentity of the current PTP Instance. This variable is reset by the current state machine.

26 *Change 11.2.19.2.4 as follows:*

27 11.2.19.2.4 rcvdPdelayRespFollowUp:

- 28 a) For full-duplex point-to-point mode: A Boolean variable that notifies the current state machine
29 when a Pdelay_Resp_Follow_Up message is received. This variable is reset by the current state
30 machine.
- 31 b) For half-duplex mode: A Boolean variable that notifies the current state machine when a
32 Pdelay_Resp_Follow_Up message is received AND its requestingPortIdentity.clockIdentity is equal
33 to the clockIdentity of the current PTP instance. This variable is reset by the current state machine.

34 *Insert a new variable after 11.2.19.2.13 as follows:*

35 **11.2.19.2.14 pdelayReqSendDisabled:** A boolean that is administratively set to TRUE if Pdelay_Req
36 messages are not transmitted by this port. The default value for this variable shall be FALSE.

11.2.19.4 State diagram

2 Replace Figure 11-9 with the following:.

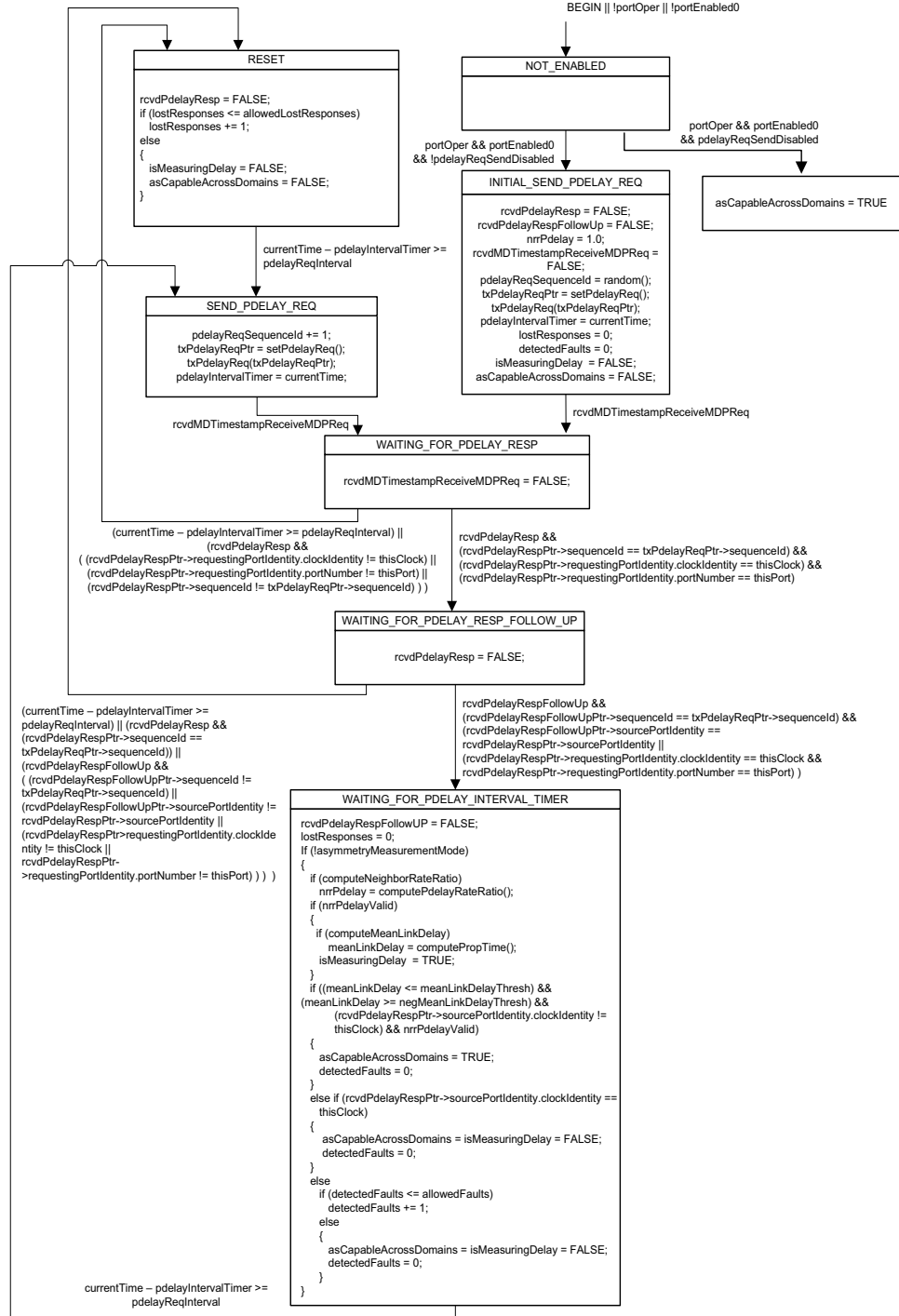


Figure 11-9—MDPdelayReq state machine

1 **Add a NOTE after Figure 11-9 as follows:**

2 NOTE—A change in the value of the variable pdelayReqSendDisabled takes effect only when portEnabled0 (see
3 11.2.19.2.12) is FALSE.

4 **11.2.20 MDPdelayResp state machine**

5 **11.2.20.2 State machine variables**

6 **Insert a new variable after 11.2.20.2.5 as follows:**

7 **11.2.20.2.6 pdelayRespSendDisabled:** A boolean that is administratively set to TRUE to indicate that
8 corresponding Pdelay_Resp and Pdelay_Resp_Follow_Up messages are not transmitted by this port in
9 response to a Pdelay_Req message. The default value for this variable shall be FALSE.

10 **11.2.20.4 State diagram**

11 **Replace Figure 11-10 with the following:**

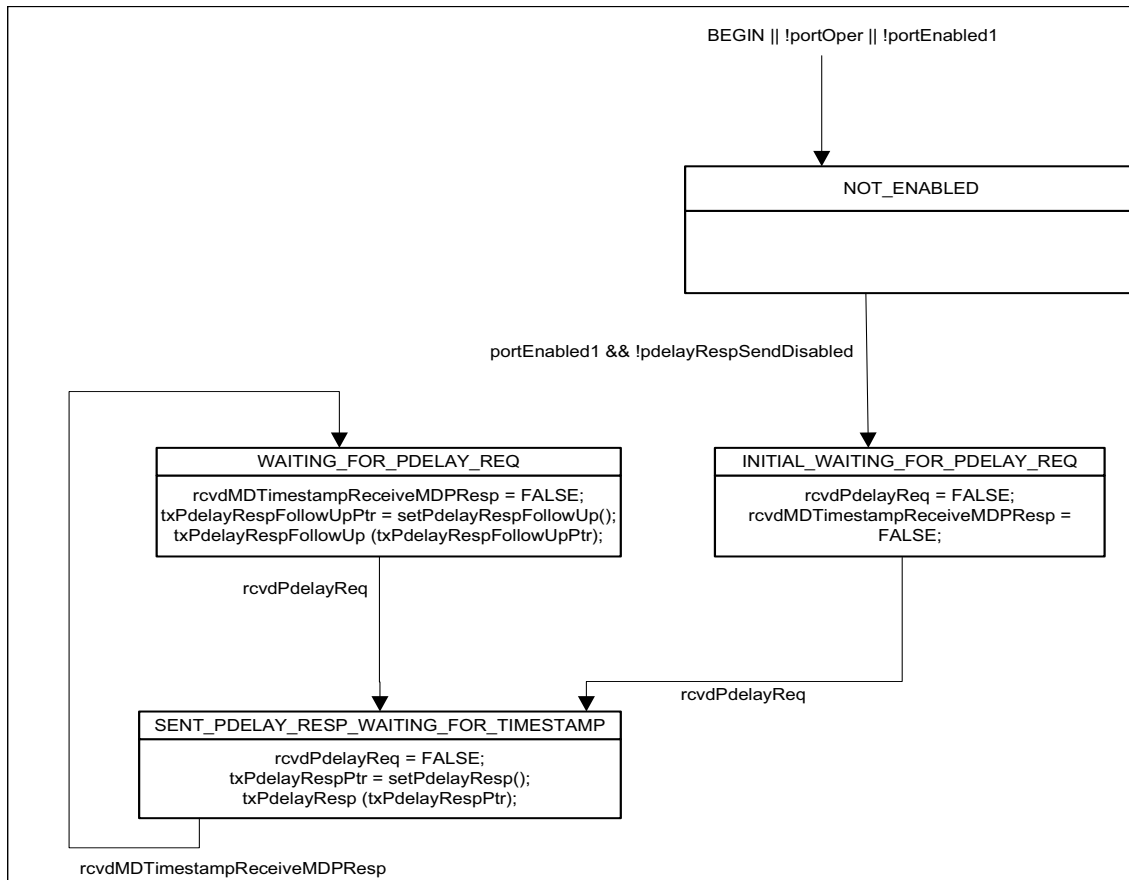


Figure 11-10—MDPdelayResp state machine

12 **Add a NOTE after Figure 11-10 as follows:**

1 NOTE—A change in the value of the variable pdelayRespSendDisabled takes effect only when portEnabled1 (see
2 11.2.20.2.5) is FALSE.

3 11.4 Message formats

4 11.4.2 Header

5 11.4.2.4 domainNumber (UInteger8)

6 *Change the paragraph in 11.4.2.4 as follows:*

7 For full-duplex, Pdelay_Req, Pdelay_Resp, and Pdelay_Resp_Follow_Up messages shall be transmitted~~The~~
8 with domainNumber ~~for Pdelay_Req, Pdelay_Resp, and Pdelay_Resp_Follow_Up messages shall be set to~~
9 0. For HDE, see 19.2.19 and 19.2.20. The domainNumber for all other PTP messages is as specified in
10 10.6.2.2.6.

1 14. Timing and synchronization management

2 14.10 Port Parameter Data Set (portDS)

3 14.10.1 General

4 *Insert the following items after the final item of Table 14-9:*

Table 14-9—portDS

Name	Data type	Operations supported ^a	References
pdelayReqSendDisabled	Boolean	RW	14.10.61
pdelayRespSendDisabled	Boolean	RW	14.10.62

^a R = Read only access; RW = Read/write access.

5 14.10.24 InitialLogPdelayReqInterval

6 *Change the first paragraph in 14.10.24 as follows:*

7 For full-duplex [and half-duplex](#) IEEE [Std](#) 802.3 media and for CSN media that use the peer-to-peer delay
 8 mechanism to measure path delay (see 16.4.3.2), the value is the logarithm to base 2 of the Pdelay_Req
 9 message transmission interval used when:

10 14.10.25 currentLogPdelayReqInterval

11 *Change the first paragraph in 14.10.25 as follows:*

12 For full-duplex [and half-duplex](#) IEEE [Std](#) 802.3 media and for CSN media that use the peer-to-peer delay
 13 mechanism to measure path delay (see 16.4.3.2), the value is the logarithm to the base 2 of the current
 14 Pdelay_Req message transmission interval (see 11.5.2.2).

15 14.10.33 currentComputeNeighborRateRatio

16 *Change the first paragraph in 14.10.33 as follows:*

17 For full-duplex [and half-duplex](#) IEEE [Std](#) 802.3 media and for CSN media that use the peer-to-peer delay
 18 mechanism to measure path delay (see 16.4.3.2), the value is the current value of
 19 computeNeighborRateRatio.

20 14.10.37 currentComputeMeanLinkDelay

21 *Change the first paragraph in 14.10.37 as follows:*

22 For full-duplex [and half-duplex](#) IEEE [Std](#) 802.3 media and for CSN media that use the peer-to-peer delay
 23 mechanism to measure path delay (see 16.4.3.2), the value is the current value of computeMeanLinkDelay.

1 14.10.54 pdelayTruncatedTimestampsArray**2 Change the first paragraph in 14.10.54 as follows:**

3 For full-duplex and half-duplex IEEE Std 802.3 media and for CSN media that use the peer-to-peer delay
4 mechanism to measure path delay (see 16.4.3.2), the values of the four elements of this array are as
5 described in Table 14-9. For all other media, the values are zero. Array elements 0, 1, 2, and 3 correspond to
6 the timestamps t1, t2, t3, and t4, respectively, in Figure 11-1 and are expressed in units of 2–16 ns (i.e., the
7 value of each array element is equal to the remainder obtained upon dividing the respective timestamp,
8 expressed in units of 2–16 ns, by 248). At any given time, the timestamp values stored in the array are for the
9 same, and most recently completed, peer delay message exchange.

10 Insert 14.10.61 and 14.10.62 as follows:**11 14.10.61 pdelayReqSendDisabled**

12 The value is equal to the value of the per-PTP Port variable pdelayReqSendDisabled (see 11.2.19.2.14). If its
13 value is TRUE, Pdelay_Req messages are not transmitted by the PTP Port. The default value for this
14 variable shall be FALSE.

15 14.10.62 pdelayRespSendDisabled

16 The value is equal to the value of the per-PTP Port variable pdelayRespSendDisabled (see 11.2.20.2.6). If its
17 value is TRUE, Pdelay_Resp and Pdelay_Resp_Follow_Up messages are not transmitted by the PTP Port in
18 response to Pdelay_Req message. The default value for this variable shall be FALSE.

17. YANG Data Model

17.2 IEEE 802.1AS YANG models

Replace Figure 17-3 with the following:

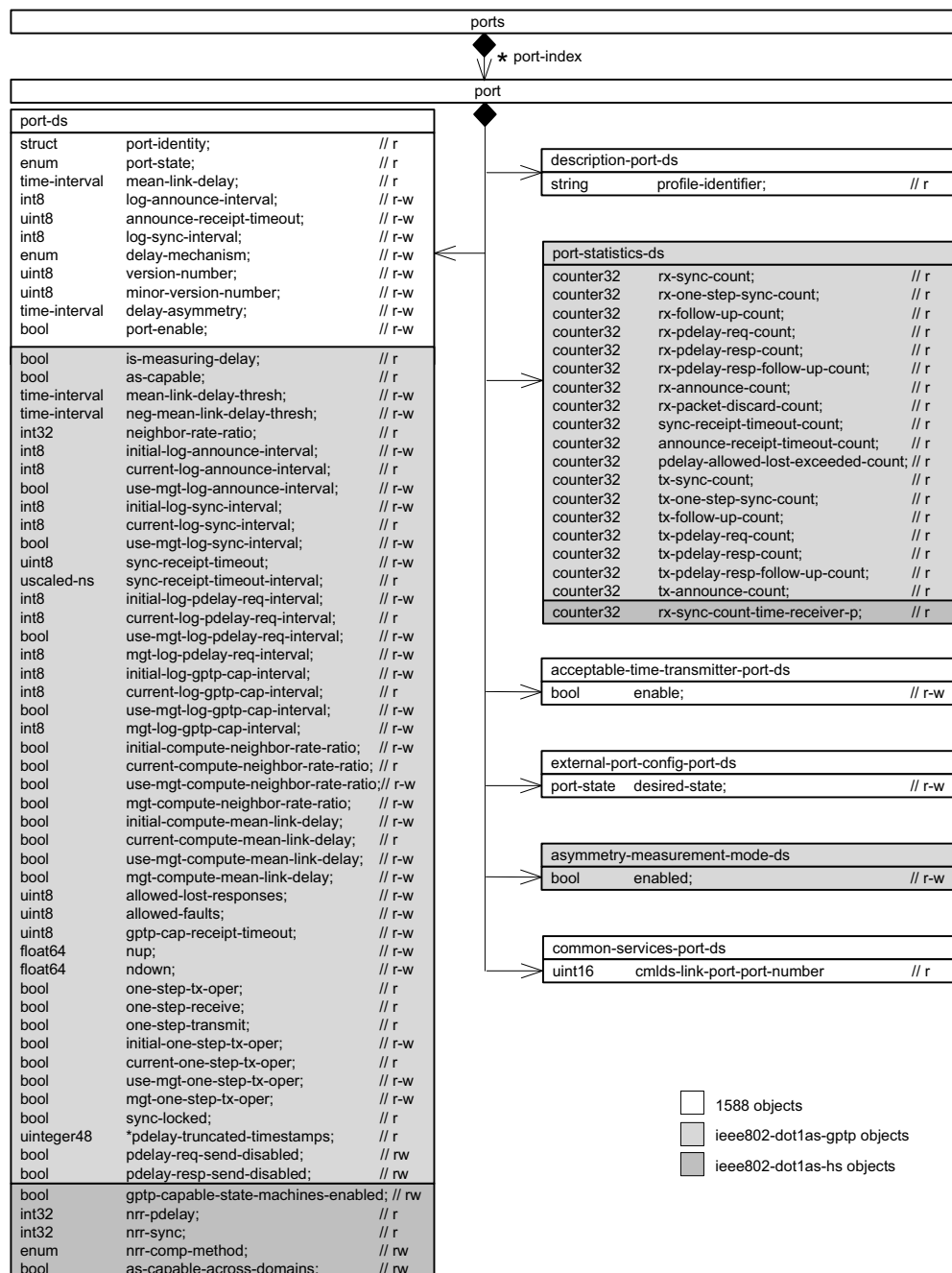


Figure 17-3—PTP Port detail

1 17.5 YANG schema tree definitions**2 17.5.1 ITree diagram for ieee802-dot1as-gptp.yang****3 *Replace the tree diagram for ieee802-dot1as-gptp.yang as follows:***

```

4 module: ieee802-dot1as-gptp
5
6   augment /ptp-tt:ptp/ptp-tt:instances/ptp-tt:instance/ptp-tt:default-ds:
7     +--ro gm-capable?                boolean
8     +--ro current-utc-offset?        int16
9     +--ro current-utc-offset-valid?  boolean
10    +--ro leap59?                    boolean
11    +--ro leap61?                    boolean
12    +--ro time-traceable?             boolean
13    +--ro frequency-traceable?        boolean
14    +--ro ptp-timescale?              boolean
15    +--ro time-source?                identityref
16  augment /ptp-tt:ptp/ptp-tt:instances/ptp-tt:instance/ptp-tt:current-ds:
17    +--ro last-gm-phase-change?        scaled-ns
18    +--ro last-gm-freq-change?         float64
19    +--ro gm-timebase-indicator?       uint16
20    +--ro gm-change-count?             yang:counter32
21    +--ro time-of-last-gm-change?      yang:timestamp
22    +--ro time-of-last-phase-change?   yang:timestamp
23    +--ro time-of-last-freq-change?    yang:timestamp
24  augment /ptp-tt:ptp/ptp-tt:instances/ptp-tt:instance/ptp-tt:parent-ds:
25    +--ro cumulative-rate-ratio?      int32
26  augment /ptp-tt:ptp/ptp-tt:instances/ptp-tt:instance/ptp-tt:ports/ptp-tt:port/
27  ptp-tt:port-ds:
28    +--ro is-measuring-delay?          boolean
29    +--ro as-capable?                  boolean
30    +--rw mean-link-delay-thresh?      ptp-tt:time-interval
31    +--rw neg-mean-link-delay-thresh?  ptp-tt:time-interval
32    +--ro neighbor-rate-ratio?         int32
33    +--rw initial-log-announce-interval? int8
34    +--ro current-log-announce-interval? int8
35    +--rw use-mgt-log-announce-interval? boolean
36    +--rw initial-log-sync-interval?    int8
37    +--ro current-log-sync-interval?    int8
38    +--rw use-mgt-log-sync-interval?    boolean
39    +--rw sync-receipt-timeout?         uint8
40    +--ro sync-receipt-timeout-interval? uscaled-ns
41    +--rw initial-log-pdelay-req-interval? int8
42    +--ro current-log-pdelay-req-interval? int8
43    +--rw use-mgt-log-pdelay-req-interval? boolean
44    +--rw mgt-log-pdelay-req-interval?   int8
45    +--rw initial-log-gptp-cap-interval? int8
46    +--ro current-log-gptp-cap-interval? int8
47    +--rw use-mgt-log-gptp-cap-interval? boolean
48    +--rw mgt-log-gptp-cap-interval?     int8
49    +--rw initial-compute-neighbor-rate-ratio? boolean
50    +--ro current-compute-neighbor-rate-ratio? boolean
51    +--rw use-mgt-compute-neighbor-rate-ratio? boolean
52    +--rw mgt-compute-neighbor-rate-ratio? boolean
53    +--rw initial-compute-mean-link-delay? boolean
54    +--ro current-compute-mean-link-delay? boolean
55    +--rw use-mgt-compute-mean-link-delay? boolean

```

```

1  +--rw mgt-compute-mean-link-delay?          boolean
2  +--rw allowed-lost-responses?                uint8
3  +--rw allowed-faults?                       uint8
4  +--rw gptp-cap-receipt-timeout?             uint8
5  +--rw nup?                                  float64
6  +--rw ndown?                                float64
7  +--ro one-step-tx-oper?                     boolean
8  +--ro one-step-receive?                    boolean
9  +--ro one-step-transmit?                   boolean
10 +--rw initial-one-step-tx-oper?             boolean
11 +--ro current-one-step-tx-oper?             boolean
12 +--rw use-mgt-one-step-tx-oper?             boolean
13 +--rw mgt-one-step-tx-oper?                 boolean
14 +--ro sync-locked?                          boolean
15 +--ro pdelay-truncated-timestamps*          uinteger48
16 +--rw pdelay-req-send-disabled?             boolean
17 +--rw pdelay-resp-send-disabled?            boolean
18 augment /ptp-tt:ptp/ptp-tt:instances/ptp-tt:instance/ptp-tt:ports/ptp-tt:port:
19   +--rw port-statistics-ds
20     +--ro rx-sync-count?                    yang:counter32
21     +--ro rx-one-step-sync-count?           yang:counter32
22     +--ro rx-follow-up-count?               yang:counter32
23     +--ro rx-pdelay-req-count?              yang:counter32
24     +--ro rx-pdelay-resp-count?             yang:counter32
25     +--ro rx-pdelay-resp-follow-up-count?   yang:counter32
26     +--ro rx-announce-count?                yang:counter32
27     +--ro rx-packet-discard-count?          yang:counter32
28     +--ro sync-receipt-timeout-count?       yang:counter32
29     +--ro announce-receipt-timeout-count?   yang:counter32
30     +--ro pdelay-allowed-lost-exceeded-count? yang:counter32
31     +--ro tx-sync-count?                    yang:counter32
32     +--ro tx-one-step-sync-count?           yang:counter32
33     +--ro tx-follow-up-count?               yang:counter32
34     +--ro tx-pdelay-req-count?              yang:counter32
35     +--ro tx-pdelay-resp-count?             yang:counter32
36     +--ro tx-pdelay-resp-follow-up-count?   yang:counter32
37     +--ro tx-announce-count?                yang:counter32
38 augment /ptp-tt:ptp/ptp-tt:instances/ptp-tt:instance/ptp-tt:ports/ptp-tt:port:
39   +--rw asymmetry-measurement-mode-ds
40   +--rw enabled? boolean
41   augment /ptp-tt:ptp/ptp-tt:common-services/ptp-tt:cmls/ptp-tt:ports/ptp-
42 tt:port/ptp-tt:link-port-ds:
43     +--ro cmls-link-port-enabled?          boolean
44     +--ro is-measuring-delay?               boolean
45     +--ro as-capable-across-domains?        boolean
46     +--rw mean-link-delay-thresh?           ptp-tt:time-interval
47     +--rw neg-mean-link-delay-thresh?       ptp-tt:time-interval
48     +--rw initial-log-pdelay-req-interval?  int8
49     +--ro current-log-pdelay-req-interval?  int8
50     +--rw use-mgt-log-pdelay-req-interval?  boolean
51     +--rw mgt-log-pdelay-req-interval?      int8
52     +--rw initial-compute-neighbor-rate-ratio? boolean
53     +--ro current-compute-neighbor-rate-ratio? boolean
54     +--rw use-mgt-compute-neighbor-rate-ratio? boolean
55     +--rw mgt-compute-neighbor-rate-ratio?  boolean
56     +--rw initial-compute-mean-link-delay?  boolean
57     +--ro current-compute-mean-link-delay?  boolean
58     +--rw use-mgt-compute-mean-link-delay?  boolean
59     +--rw mgt-compute-mean-link-delay?      boolean

```

```

1  +--rw allowed-lost-responses?          uint8
2  +--rw allowed-faults?                  uint8
3  +--ro pdelay-truncated-timestamps*      integer48
4  augment /ptp-tt:ptp/ptp-tt:common-services/ptp-tt:cmlDs/ptp-tt:ports/ptp-
5  tt:port:
6  +--rw port-statistics-ds
7      +--ro rx-pdelay-req-count?          yang:counter32
8      +--ro rx-pdelay-resp-count?        yang:counter32
9      +--ro rx-pdelay-resp-follow-up-count? yang:counter32
10     +--ro rx-packet-discard-count?      yang:counter32
11     +--ro pdelay-allowed-lost-exceeded-count? yang:counter32
12     +--ro tx-pdelay-req-count?          yang:counter32
13     +--ro tx-pdelay-resp-count?        yang:counter32
14     +--ro tx-pdelay-resp-follow-up-count? yang:counter32
15     augment /ptp-tt:ptp/ptp-tt:common-services/ptp-tt:cmlDs/ptp-tt:ports/ptp-
16     tt:port:
17     +--rw asymmetry-measurement-mode-ds
18     +--rw enabled?      boolean
19

```

17.6 YANG modules^{1 2}

17.6.1 Module `ieee802-dot1as-gtp.yang`

Replace the YANG module for `ieee802-dot1as-gtp.yang` as follows:

```

23 module ieee802-dot1as-gtp {
24   yang-version "1.1";
25   namespace urn:ieee:std:802.1AS:yang:ieee802-dot1as-gtp;
26   prefix dot1as-gtp;
27
28   import ietf-yang-types {
29     prefix yang;
30   }
31   import ieee1588-ptp-tt {
32     prefix ptp-tt;
33   }
34
35   organization
36     "IEEE 802.1 Working Group";
37   contact
38     "WG-URL: http://ieee802.org/1/
39     WG-EMail: stds-802-1-l@ieee.org
40
41     Contact: IEEE 802.1 Working Group Chair
42             Postal: C/O IEEE 802.1 Working Group
43             IEEE Standards Association
44             445 Hoes Lane
45             Piscataway, NJ 08854
46             USA
47
48     E-mail: stds-802-1-chairs@ieee.org";
49   description

```

¹Copyright release for YANG modules: Users of this standard may freely reproduce the YANG modules contained in this subclause so that they can be used for their intended purpose.

²An ASCII version of the YANG modules are attached to the PDF version of this standard, and can be obtained by Web browser from the IEEE 802.1 Website at <https://1.ieee802.org/yang-modules/>.

```

1  "Management objects that control timing and synchronization for
2  time sensitive applications, as specified in Clause 14 of
3  IEEE Std 802.1AS-2025.
4
5  Copyright (C) IEEE (2025). This version of this YANG module is
6  part of IEEE Std 802.1AS-2025; see the standard itself for full
7  legal notices.";
8
9  revision 2026-01-20 {
10   description
11     "Published as part of IEEE Std 802.1AS-2025.";
12   reference
13     "IEEE Std 802.1AS - Timing and Synchronization for
14     Time-Sensitive Applications: IEEE Std 802.1AS-2025.
15     IEEE Std 1588 - IEEE Standard for a Precision Clock
16     Synchronization Protocol for Networked Measurement and
17     Control Systems: IEEE Std 1588-2019, IEEE Std 1588g-2022,
18     IEEE Std 1588e-2024.";
19 }
20 typedef scaled-ns {
21   type string {
22     pattern "[0-9A-F]{2}(-[0-9A-F]{2}){11}";
23   }
24   description
25     "The IEEE Std 802.1AS ScaledNs type represents signed values
26     of time and time interval in units of 2^16 ns, as a signed
27     96-bit integer. Each of the 12 octets is represented as a
28     pair of hexadecimal characters, using uppercase for a letter.
29     Octets are separated by a dash character. The most
30     significant octet is first.";
31   reference
32     "6.4.3.1 of IEEE Std 802.1AS";
33 }
34 typedef unscaled-ns {
35   type string {
36     pattern "[0-9A-F]{2}(-[0-9A-F]{2}){11}";
37   }
38   description
39     "The IEEE Std 802.1AS UScaledNs type represents unsigned
40     values of time and time interval in units of 2^16 ns, as an
41     unsigned 96-bit integer. Each of the 12 octets is represented
42     as a pair of hexadecimal characters, using uppercase for a
43     letter. Octets are separated by a dash character. The most
44     significant octet is first.";
45   reference
46     "6.4.3.2 of IEEE Std 802.1AS";
47 }
48 typedef float64 {
49   type string {
50     pattern "[0-9A-F]{2}(-[0-9A-F]{2}){7}";
51   }
52   description
53     "The IEEE Std 802.1AS Float64 type represents IEEE Std 754
54     binary64. Each of the 8 octets is represented as a pair of
55     hexadecimal characters, using uppercase for a letter. Octets
56     are separated by a dash character. The most significant octet
57     is first.";
58   reference
59     "6.4.2 of IEEE Std 802.1AS";

```



```

1  }
2  typedef uinteger48 {
3      type uint64 {
4          range "0..281474976710655";
5      }
6      description
7          "48-bit unsigned integer data type.";
8      reference
9          "6.4.2 of IEEE Std 802.1AS";
10 }
11
12 augment "/ptp-tt:ptp/ptp-tt:instances/ptp-tt:instance"
13     + "/ptp-tt:default-ds" {
14     description
15         "Augment IEEE Std 1588 defaultDS.";
16     leaf gm-capable {
17         type boolean;
18         config false;
19         description
20             "The value is true if the time-aware system is capable of
21              being a Grandmaster, and false if the time-aware system is
22              not capable of being a Grandmaster.";
23         reference
24             "14.2.7 of IEEE Std 802.1AS";
25     }
26     leaf current-utc-offset {
27         when
28             "../current-utc-offset-valid='true'";
29         type int16;
30         config false;
31         description
32             "Offset from UTC (TAI - UTC). The offset is in units of
33              seconds. This leaf applies to the ClockTimeTransmitter
34              entity (i.e., local only, unrelated to a remote GM).";
35         reference
36             "14.2.8 of IEEE Std 802.1AS";
37     }
38     leaf current-utc-offset-valid {
39         type boolean;
40         config false;
41         description
42             "The value of current-utc-offset-valid shall be true if the
43              value of current-utc-offset is known to be correct,
44              otherwise it shall be false. This leaf applies to the
45              ClockTimeTransmitter entity (i.e., local only, unrelated to
46              a remote GM).";
47         reference
48             "14.2.9 of IEEE Std 802.1AS";
49     }
50     leaf leap59 {
51         type boolean;
52         config false;
53         description
54             "If the timescale is PTP, a true value for leap59 shall
55              indicate that the last minute of the current UTC day
56              contains 59 seconds. If the timescale is not PTP, the value
57              shall be false. This leaf applies to the
58              ClockTimeTransmitter entity (i.e., local only, unrelated to
59              a remote GM).";

```

```

1      reference
2      "14.2.10 of IEEE Std 802.1AS";
3  }
4  leaf leap61 {
5      type boolean;
6      config false;
7      description
8          "If the timescale is PTP, a true value for leap61 shall
9          indicate that the last minute of the current UTC day
10         contains 61 seconds. If the timescale is not PTP, the value
11         shall be false. This leaf applies to the
12         ClockTimeTransmitter entity (i.e., local only, unrelated to
13         a remote GM).";
14     reference
15         "14.2.11 of IEEE Std 802.1AS";
16 }
17 leaf time-traceable {
18     type boolean;
19     config false;
20     description
21         "The value of time-traceable shall be true if the timescale
22         is traceable to a primary reference; otherwise, the value
23         shall be false. This leaf applies to the
24         ClockTimeTransmitter entity (i.e., local only, unrelated to
25         a remote GM).";
26     reference
27         "14.2.12 of IEEE Std 802.1AS";
28 }
29 leaf frequency-traceable {
30     type boolean;
31     config false;
32     description
33         "The value of frequency-traceable shall be true if the
34         frequency determining the timescale is traceable to a
35         primary reference; otherwise, the value shall be false.
36         This leaf applies to the ClockTimeTransmitter entity
37         (i.e., local only, unrelated to a remote GM).";
38     reference
39         "14.2.13 of IEEE Std 802.1AS";
40 }
41 leaf ptp-timescale {
42     type boolean;
43     config false;
44     description
45         "If ptp-timescale is true, the timescale of the
46         ClockTimeTransmitter entity is PTP, which is the elapsed
47         time since the PTP epoch measured using the second defined
48         by International Atomic Time (TAI). If ptp-timescale is
49         false, the timescale of the ClockTimeTransmitter entity is
50         ARB, which is the elapsed time since an arbitrary epoch.
51         This leaf applies to the ClockTimeTransmitter entity
52         (i.e., local only, unrelated to a remote GM).";
53     reference
54         "14.2.14 of IEEE Std 802.1AS";
55 }
56 leaf time-source {
57     type identityref {
58         base ptp-tt:time-source;
59     }

```

```

1      config false;
2      description
3          "The source of time used by the Grandmaster Clock. This leaf
4          applies to the ClockTimeTransmitter entity (i.e., local
5          only, unrelated to a remote GM).";
6      reference
7          "14.2.15 of IEEE Std 802.1AS";
8  }
9  }
10
11  augment "/ptp-tt:ptp/ptp-tt:instances/ptp-tt:instance"
12      + "/ptp-tt:current-ds" {
13      description
14          "Augment IEEE Std 1588 currentDS.";
15      leaf last-gm-phase-change {
16          type scaled-ns;
17          config false;
18          description
19              "Phase change that occurred on the most recent change in
20              either the Grandmaster PTP Instance or
21              gm-timebase-indicator leaf.";
22          reference
23              "14.3.4 of IEEE Std 802.1AS";
24      }
25      leaf last-gm-freq-change {
26          type float64;
27          config false;
28          description
29              "Frequency change that occurred on the most recent change in
30              either the Grandmaster PTP Instance or
31              gm-timebase-indicator leaf.";
32          reference
33              "14.3.5 of IEEE Std 802.1AS";
34      }
35      leaf gm-timebase-indicator {
36          type uint16;
37          config false;
38          description
39              "The timeBaseIndicator of the current Grandmaster PTP
40              Instance.";
41          reference
42              "14.3.6 of IEEE Std 802.1AS";
43      }
44      leaf gm-change-count {
45          type yang:counter32;
46          config false;
47          description
48              "This statistics counter tracks the number of times the
49              Grandmaster PTP Instance has changed in a gPTP domain.";
50          reference
51              "14.3.7 of IEEE Std 802.1AS";
52      }
53      leaf time-of-last-gm-change {
54          type yang:timestamp;
55          config false;
56          description
57              "System time when the most recent Grandmaster Clock change
58              occurred in a gPTP domain. This leaf's type is YANG
59              timestamp, which is based on system time. System time is an

```

```

1      unsigned integer in units of 10 milliseconds, using an
2      epoch defined by the implementation (typically time of
3      boot-up).";
4      reference
5      "14.3.8 of IEEE Std 802.1AS";
6  }
7  leaf time-of-last-phase-change {
8      type yang:timestamp;
9      config false;
10     description
11         "System time when the most recent change in Grandmaster
12         Clock phase occurred. This leaf's type is YANG timestamp,
13         which is based on system time. System time is an unsigned
14         integer in units of 10 milliseconds, using an epoch defined
15         by the implementation (typically time of boot-up).";
16     reference
17         "14.3.9 of IEEE Std 802.1AS";
18 }
19 leaf time-of-last-freq-change {
20     type yang:timestamp;
21     config false;
22     description
23         "System time when the most recent change in Grandmaster
24         Clock frequency occurred. This leaf's type is YANG
25         timestamp, which is based on system time. System time is an
26         unsigned integer in units of 10 milliseconds, using an
27         epoch defined by the implementation (typically time of
28         boot-up).";
29     reference
30         "14.3.10 of IEEE Std 802.1AS";
31 }
32 }
33
34 augment "/ptp-tt:ptp/ptp-tt:instances/ptp-tt:instance"
35     + "/ptp-tt:parent-ds" {
36     description
37         "Augment IEEE Std 1588 parentDS.";
38     leaf cumulative-rate-ratio {
39         type int32;
40         config false;
41         description
42             "Estimate of the ratio of the frequency of the Grandmaster
43             Clock to the frequency of the LocalClock entity of this PTP
44             Instance. cumulative-rate-ratio is expressed as the
45             fractional frequency offset multiplied by 2^41, i.e., the
46             quantity (rateRatio - 1.0)(2^41).";
47         reference
48             "14.4.3 of IEEE Std 802.1AS";
49     }
50 }
51
52 augment
53     "/ptp-tt:ptp"+
54     "/ptp-tt:instances"+
55     "/ptp-tt:instance"+
56     "/ptp-tt:ports"+
57     "/ptp-tt:port"+
58     "/ptp-tt:port-ds" {
59     description

```

```

1      "Augment IEEE Std 1588 portDS.
2
3      14.10.4 of IEEE Std 802.1AS specifies ptpPortEnabled
4      (ptp-port-enabled), which is provided in YANG as the
5      semantically equivalent node in ieee1588-ptp-tt named
6      port-enable (in port-ds).
7
8      14.10.16 of IEEE Std 802.1AS specifies
9      mgtSettableLogAnnounceInterval (mgt-log-announce-interval),
10     which is provided in YANG as the semantically equivalent node in
11     ieee1588-ptp-tt named log-announce-interval (in port-ds).
12     In the context of IEEE Std 802.1AS, log-announce-interval
13     cannot be used unless use-mgt-log-announce-interval is true.
14
15     14.10.21 of IEEE Std 802.1AS specifies
16     mgtSettableLogSyncInterval (mgt-log-sync-interval), which is
17     provided in YANG as the semantically equivalent node in
18     ieee1588-ptp-tt named log-sync-interval (in port-ds). In the
19     context of IEEE Std 802.1AS, log-sync-interval cannot be used
20     unless use-mgt-log-sync-interval is true.";
21 leaf is-measuring-delay {
22     type boolean;
23     config false;
24     description
25         "Boolean that is true if the port is measuring PTP Link
26         propagation delay.";
27     reference
28         "14.10.6 of IEEE Std 802.1AS";
29 }
30 leaf as-capable {
31     type boolean;
32     config false;
33     description
34         "Boolean that is true if and only if it is determined that
35         this PTP Instance and the PTP Instance at the other end of
36         the link attached to this port can interoperate with each
37         other via the IEEE Std 802.1AS protocol.";
38     reference
39         "10.2.5.1 of IEEE Std 802.1AS
40         14.10.7 of IEEE Std 802.1AS";
41 }
42 leaf mean-link-delay-thresh {
43     type ptp-tt:time-interval;
44     description
45         "Propagation time threshold for mean-link-delay, above which
46         a port is not considered capable of participating in the
47         IEEE Std 802.1AS protocol.";
48     reference
49         "14.10.9 of IEEE Std 802.1AS";
50 }
51 leaf neg-mean-link-delay-thresh {
52     type ptp-tt:time-interval;
53     description
54         "The negative propagation time threshold for
55         mean-link-delay, below which a port is not considered
56         capable of participating in the IEEE Std 802.1AS
57         protocol.";
58     reference
59         "14.10.10 of IEEE Std 802.1AS";

```

```

1      }
2      leaf neighbor-rate-ratio {
3          type int32;
4          config false;
5          description
6              "Estimate of the ratio of the frequency of the LocalClock
7              entity of the PTP Instance at the other end of the link
8              attached to this PTP Port, to the frequency of the
9              LocalClock entity of this PTP Instance. neighbor-rate-ratio
10             is expressed as the fractional frequency offset multiplied
11             by 2^41, i.e., the quantity (rateRatio - 1.0)(2^41).";
12         reference
13             "14.10.12 of IEEE Std 802.1AS";
14     }
15     leaf initial-log-announce-interval {
16         type int8;
17         description
18             "When use-mgt-log-announce-interval is false (i.e., change
19             with Signaling message), this is the logarithm to base 2 of
20             the announce interval used when the port is initialized.";
21         reference
22             "14.10.13 of IEEE Std 802.1AS";
23     }
24     leaf current-log-announce-interval {
25         type int8;
26         config false;
27         description
28             "Logarithm to base 2 of the current announce interval.";
29         reference
30             "14.10.14 of IEEE Std 802.1AS";
31     }
32     leaf use-mgt-log-announce-interval {
33         type boolean;
34         description
35             "Boolean that determines the source of the announce
36             interval. If the value is true, the announce interval
37             (current-log-announce-interval) is set equal to the value
38             of mgt-log-announce-interval. If the value is false, the
39             announce interval is determined by the
40             AnnounceIntervalSetting state machine (i.e., changed with
41             Signaling message).";
42         reference
43             "14.10.15 of IEEE Std 802.1AS";
44     }
45     leaf initial-log-sync-interval {
46         type int8;
47         description
48             "When use-mgt-log-sync-interval is false (i.e., change with
49             Signaling message), this is the logarithm to base 2 of the
50             sync interval used when the port is initialized.";
51         reference
52             "14.10.18 of IEEE Std 802.1AS";
53     }
54     leaf current-log-sync-interval {
55         type int8;
56         config false;
57         description
58             "Logarithm to base 2 of the current sync interval.";
59         reference

```

```

1      "14.10.19 of IEEE Std 802.1AS";
2  }
3  leaf use-mgt-log-sync-interval {
4      type boolean;
5      description
6          "Boolean that determines the source of the sync interval.
7           If the value is true, the sync interval
8           (current-log-sync-interval) is set equal to the value of
9           mgt-log-sync-interval. If the value is false, the sync
10          interval is determined by the SyncIntervalSetting
11          state machine (i.e., changed with Signaling message).";
12      reference
13          "14.10.20 of IEEE Std 802.1AS";
14  }
15  leaf sync-receipt-timeout {
16      type uint8;
17      description
18          "Number of sync intervals that a timeReceiver port waits
19           without receiving synchronization information, before
20           assuming that the timeTransmitter is no longer transmitting
21           synchronization information and that the BTCA needs to be
22           run, if appropriate.";
23      reference
24          "14.10.22 of IEEE Std 802.1AS";
25  }
26  leaf sync-receipt-timeout-interval {
27      type uscaled-ns;
28      config false;
29      description
30          "Time interval after which sync receipt timeout occurs if
31           time-synchronization information has not been received
32           during the interval.";
33      reference
34          "14.10.23 of IEEE Std 802.1AS";
35  }
36  leaf initial-log-pdelay-req-interval {
37      type int8;
38      description
39          "When use-mgt-log-pdelay-req-interval is false (i.e., change
40           with Signaling message), this is the logarithm to base 2 of
41           the Pdelay_Req transmit interval used when the port is
42           initialized.";
43      reference
44          "14.10.24 of IEEE Std 802.1AS";
45  }
46  leaf current-log-pdelay-req-interval {
47      type int8;
48      config false;
49      description
50          "Logarithm to base 2 of the current Pdelay_Req transmit
51           interval.";
52      reference
53          "14.10.25 of IEEE Std 802.1AS";
54  }
55  leaf use-mgt-log-pdelay-req-interval {
56      type boolean;
57      description
58          "Boolean that determines the source of the Pdelay_Req
59           transmit interval. If the value is true, the Pdelay_Req

```

```

1      transmit interval (current-log-pdelay-req-interval) is set
2      equal to the value of mgt-log-pdelay-req-interval. If the
3      value is false, the Pdelay_Req transmit interval is
4      determined by the LinkDelayIntervalSetting state machine
5      (i.e., changed with Signaling message).";
6      reference
7      "14.10.26 of IEEE Std 802.1AS";
8  }
9  leaf mgt-log-pdelay-req-interval {
10     type int8;
11     description
12         "Logarithm to base 2 of the Pdelay_Req transmit interval,
13         used if use-mgt-log-pdelay-req-interval is true. This value
14         is not used if use-mgt-log-pdelay-req-interval is false.";
15     reference
16         "14.10.27 of IEEE Std 802.1AS";
17 }
18 leaf initial-log-gtp-cap-interval {
19     type int8;
20     description
21         "When use-mgt-log-gtp-cap-interval is false (i.e., change
22         with Signaling message), this is the logarithm to base 2 of
23         the gTP capable message interval used when the port is
24         initialized.";
25     reference
26         "14.10.28 of IEEE Std 802.1AS";
27 }
28 leaf current-log-gtp-cap-interval {
29     type int8;
30     config false;
31     description
32         "Logarithm to base 2 of the current gTP capable message
33         interval.";
34     reference
35         "14.10.29 of IEEE Std 802.1AS";
36 }
37 leaf use-mgt-log-gtp-cap-interval {
38     type boolean;
39     description
40         "Boolean that determines the source of the gTP capable
41         message interval. If the value is true, the gTP capable
42         message interval (current-log-gtp-cap-interval) is set
43         equal to the value of mgt-gtp-cap-req-interval. If the
44         value is false, the gTP capable message interval is
45         determined by the GtpCapableMessageIntervalSetting state
46         machine (i.e., changed with Signaling message).";
47     reference
48         "14.10.30 of IEEE Std 802.1AS";
49 }
50 leaf mgt-log-gtp-cap-interval {
51     type int8;
52     description
53         "Logarithm to base 2 of the gTP capable message interval,
54         used if use-mgt-log-gtp-cap-interval is true. This value
55         is not used if use-mgt-log-pdelay-req-interval is false.";
56     reference
57         "14.10.31 of IEEE Std 802.1AS";
58 }
59 leaf initial-compute-neighbor-rate-ratio {

```



```

1      type boolean;
2      description
3          "When use-mgt-compute-neighbor-rate-ratio is false
4              (i.e., change with Signaling message), this is the initial
5              value of computeNeighborRateRatio.";
6      reference
7          "14.10.32 of IEEE Std 802.1AS";
8  }
9  leaf current-compute-neighbor-rate-ratio {
10     type boolean;
11     config false;
12     description
13         "Current value of computeNeighborRateRatio.";
14     reference
15         "14.10.33 of IEEE Std 802.1AS";
16 }
17 leaf use-mgt-compute-neighbor-rate-ratio {
18     type boolean;
19     description
20         "Boolean that determines the source of
21             computeNeighborRateRatio. If the value is true,
22             computeNeighborRateRatio is set equal to the value of
23             mgt-compute-neighbor-rate-ratio. If the value is false,
24             computeNeighborRateRatio is determined by the
25             LinkDelayIntervalSetting state machine (i.e., changed with
26             Signaling message).";
27     reference
28         "14.10.34 of IEEE Std 802.1AS";
29 }
30 leaf mgt-compute-neighbor-rate-ratio {
31     type boolean;
32     description
33         "Value of computeNeighborRateRatio, used if
34             use-mgt-compute-neighbor-rate-ratio is true. This value is
35             not used if use-mgt-compute-neighbor-rate-ratio is false.";
36     reference
37         "14.10.35 of IEEE Std 802.1AS";
38 }
39 leaf initial-compute-mean-link-delay {
40     type boolean;
41     description
42         "When use-mgt-compute-mean-link-delay is false (i.e., change
43             with Signaling message), this is the initial value of
44             computeMeanLinkDelay.";
45     reference
46         "14.10.36 of IEEE Std 802.1AS";
47 }
48 leaf current-compute-mean-link-delay {
49     type boolean;
50     config false;
51     description
52         "Current value of computeMeanLinkDelay.";
53     reference
54         "14.10.37 of IEEE Std 802.1AS";
55 }
56 leaf use-mgt-compute-mean-link-delay {
57     type boolean;
58     description
59         "Boolean that determines the source of computeMeanLinkDelay."

```

```

1      If the value is true, computeMeanLinkDelay is set equal to
2      the value of mgt-compute-mean-link-delay. If the value is
3      false, computeMeanLinkDelay is determined by the
4      LinkDelayIntervalSetting state machine (i.e., changed with
5      Signaling message).";
6      reference
7      "14.10.38 of IEEE Std 802.1AS";
8  }
9  leaf mgt-compute-mean-link-delay {
10     type boolean;
11     description
12         "Value of computeMeanLinkDelay, used if
13         use-mgt-compute-mean-link-delay is true. This value is not
14         used if use-mgt-compute-mean-link-delay is false.";
15     reference
16         "14.10.39 of IEEE Std 802.1AS";
17 }
18 leaf allowed-lost-responses {
19     type uint8;
20     description
21         "Number of Pdelay_Req messages for which a valid response is
22         not received, above which a port is considered to not be
23         exchanging peer delay messages with its neighbor.";
24     reference
25         "14.10.40 of IEEE Std 802.1AS";
26 }
27 leaf allowed-faults {
28     type uint8;
29     description
30         "Number of faults above which asCapable is set to false.";
31     reference
32         "14.10.41 of IEEE Std 802.1AS";
33 }
34 leaf gtp-cap-receipt-timeout {
35     type uint8;
36     description
37         "Number of transmission intervals that a port waits without
38         receiving the gPTP-capable TLV, before assuming that the
39         neighbor port is no longer invoking the gPTP protocol.";
40     reference
41         "14.10.42 of IEEE Std 802.1AS";
42 }
43 leaf nup {
44     type float64;
45     description
46         "For an OLT port of an IEEE Std 802.3 EPON link, this value
47         is the effective index of refraction for the EPON upstream
48         wavelength light of the optical path.";
49     reference
50         "14.10.44 of IEEE Std 802.1AS";
51 }
52 leaf ndown {
53     type float64;
54     description
55         "For an OLT port of an IEEE 802.3 EPON link, this value is
56         the effective index of refraction for the EPON downstream
57         wavelength light of the optical path.";
58     reference
59         "14.10.45 of IEEE Std 802.1AS";

```

```

1      }
2      leaf one-step-tx-oper {
3          type boolean;
4          config false;
5          description
6              "This value is true if the port is sending one-step Sync
7              messages, and false if the port is sending two-step Sync
8              and Follow_Up messages.";
9          reference
10             "14.10.46 of IEEE Std 802.1AS";
11     }
12     leaf one-step-receive {
13         type boolean;
14         config false;
15         description
16             "This value is true if the port is capable of receiving and
17             processing one-step Sync messages.";
18         reference
19             "14.10.47 of IEEE Std 802.1AS";
20     }
21     leaf one-step-transmit {
22         type boolean;
23         config false;
24         description
25             "This value is true if the port is capable of transmitting
26             one-step Sync messages.";
27         reference
28             "14.10.48 of IEEE Std 802.1AS";
29     }
30     leaf initial-one-step-tx-oper {
31         type boolean;
32         description
33             "When use-mgt-one-step-tx-oper is false (i.e., change with
34             Signaling message), this is the initial value of
35             current-one-step-tx-oper.";
36         reference
37             "14.10.49 of IEEE Std 802.1AS";
38     }
39     leaf current-one-step-tx-oper {
40         type boolean;
41         config false;
42         description
43             "This value is true if the port is configured to transmit
44             one-step Sync messages, either via management
45             (mgt-one-step-tx-oper) or Signaling. If both
46             current-one-step-tx-oper and one-step-transmit are true,
47             the port transmits one-step Sync messages
48             (i.e., one-step-tx-oper true).";
49         reference
50             "14.10.50 of IEEE Std 802.1AS";
51     }
52     leaf use-mgt-one-step-tx-oper {
53         type boolean;
54         description
55             "Boolean that determines the source of
56             current-one-step-tx-oper. If the value is true,
57             current-one-step-tx-oper is set equal to the value of
58             mgt-one-step-tx-oper. If the value is false,
59             current-one-step-tx-oper is determined by the

```

```

1      OneStepTxOperSetting state machine (i.e., changed with
2      Signaling message).";
3      reference
4      "14.10.51 of IEEE Std 802.1AS";
5  }
6  leaf mgt-one-step-tx-oper {
7      type boolean;
8      description
9      "If use-mgt-one-step-tx-oper is true,
10      current-one-step-tx-oper is set equal to this value. This
11      value is not used if use-mgt-one-step-tx-oper is false.";
12      reference
13      "14.10.52 of IEEE Std 802.1AS";
14  }
15  leaf sync-locked {
16      type boolean;
17      config false;
18      description
19      "This value is true if the port will transmit a Sync as soon
20      as possible after the timeReceiver port receives a Sync
21      message.";
22      reference
23      "14.10.53 of IEEE Std 802.1AS";
24  }
25  leaf-list pdelay-truncated-timestamps {
26      type uinteger48;
27      config false;
28      description
29      "For full-duplex IEEE Std 802.3 media, and CSN media that
30      use the peer-to-peer delay mechanism to measure path delay,
31      the values of the four elements of this leaf-list
32      correspond to the timestamps t1, t2, t3, and t4, listed in
33      that order. Each timestamp is expressed in units of
34      2-16 ns (i.e., the value of each array element is equal to
35      the remainder obtained upon dividing the respective
36      timestamp, expressed in units of 2-16 ns, by 248).
37      At any given time, the timestamp values stored in the array
38      are for the same, and most recently completed, peer delay
39      message exchange. For each timestamp, only 48-bits are
40      valid (the upper 16-bits are always zero).";
41      reference
42      "14.10.54 of IEEE Std 802.1AS";
43  }
44  leaf pdelay-req-send-disabled {
45      type boolean;
46      description
47      "A boolean that is administratively set to TRUE
48      if Pdelay_Req messages are not transmitted by this port.
49      The default value for this variable shall be FALSE.";
50      reference
51      "14.10.61 of IEEE Std 802.1AS";
52  }
53  leaf pdelay-resp-send-disabled {
54      type boolean;
55      description
56      "A boolean that is administratively set to TRUE
57      if Pdelay_Resp messages are not transmitted by this port.
58      The default value for this variable shall be FALSE.";
59      reference

```

```

1      "14.10.62 of IEEE Std 802.1AS";
2  }
3  }
4
5  augment
6    "/ptp-tt:ptp"+
7    "/ptp-tt:instances"+
8    "/ptp-tt:instance"+
9    "/ptp-tt:ports"+
10   "/ptp-tt:port" {
11   description
12     "Augment to add port-statistics-ds to IEEE Std 1588 PTP
13     Port.";
14   container port-statistics-ds {
15     description
16       "Provides counters associated with the port of the PTP
17       Instance.";
18     reference
19       "14.12 of IEEE Std 802.1AS";
20     leaf rx-sync-count {
21       type yang:counter32;
22       config false;
23       description
24         "Counter that increments every time synchronization
25         information is received.";
26       reference
27         "14.12.2 of IEEE Std 802.1AS";
28     }
29     leaf rx-one-step-sync-count {
30       type yang:counter32;
31       config false;
32       description
33         "Counter that increments every time a one-step Sync
34         message is received.";
35       reference
36         "14.12.3 of IEEE Std 802.1AS";
37     }
38     leaf rx-follow-up-count {
39       type yang:counter32;
40       config false;
41       description
42         "Counter that increments every time a Follow_Up message is
43         received.";
44       reference
45         "14.12.4 of IEEE Std 802.1AS";
46     }
47     leaf rx-pdelay-req-count {
48       type yang:counter32;
49       config false;
50       description
51         "Counter that increments every time a Pdelay_Req message
52         is received.";
53       reference
54         "14.12.5 of IEEE Std 802.1AS";
55     }
56     leaf rx-pdelay-resp-count {
57       type yang:counter32;
58       config false;
59       description

```

```

1      "Counter that increments every time a Pdelay_Resp message
2      is received.";
3      reference
4      "14.12.6 of IEEE Std 802.1AS";
5    }
6    leaf rx-pdelay-resp-follow-up-count {
7      type yang:counter32;
8      config false;
9      description
10     "Counter that increments every time a
11     Pdelay_Resp_Follow_Up message is received.";
12     reference
13     "14.12.7 of IEEE Std 802.1AS";
14   }
15   leaf rx-announce-count {
16     type yang:counter32;
17     config false;
18     description
19     "Counter that increments every time an Announce message is
20     received.";
21     reference
22     "14.12.8 of IEEE Std 802.1AS";
23   }
24   leaf rx-packet-discard-count {
25     type yang:counter32;
26     config false;
27     description
28     "Counter that increments every time a PTP message of the
29     respective PTP Instance is discarded.";
30     reference
31     "14.12.9 of IEEE Std 802.1AS";
32   }
33   leaf sync-receipt-timeout-count {
34     type yang:counter32;
35     config false;
36     description
37     "Counter that increments every time a sync receipt timeout
38     occurs.";
39     reference
40     "14.12.10 of IEEE Std 802.1AS";
41   }
42   leaf announce-receipt-timeout-count {
43     type yang:counter32;
44     config false;
45     description
46     "Counter that increments every time an announce receipt
47     timeout occurs.";
48     reference
49     "14.12.11 of IEEE Std 802.1AS";
50   }
51   leaf pdelay-allowed-lost-exceeded-count {
52     type yang:counter32;
53     config false;
54     description
55     "Counter that increments every time the value of the
56     variable lostResponses exceeds the value of the variable
57     allowedLostResponses, in the RESET state of the
58     MDPdelayReq state machine.";
59     reference

```

```

1      "14.12.12 of IEEE Std 802.1AS";
2  }
3  leaf tx-sync-count {
4      type yang:counter32;
5      config false;
6      description
7          "Counter that increments every time synchronization
8          information is transmitted.";
9      reference
10         "14.12.13 of IEEE Std 802.1AS";
11 }
12 leaf tx-one-step-sync-count {
13     type yang:counter32;
14     config false;
15     description
16         "Counter that increments every time a one-step Sync
17         message is transmitted.";
18     reference
19         "14.12.14 of IEEE Std 802.1AS";
20 }
21 leaf tx-follow-up-count {
22     type yang:counter32;
23     config false;
24     description
25         "Counter that increments every time a Follow_Up message is
26         transmitted.";
27     reference
28         "14.12.15 of IEEE Std 802.1AS";
29 }
30 leaf tx-pdelay-req-count {
31     type yang:counter32;
32     config false;
33     description
34         "Counter that increments every time a Pdelay_Req message
35         is transmitted.";
36     reference
37         "14.12.16 of IEEE Std 802.1AS";
38 }
39 leaf tx-pdelay-resp-count {
40     type yang:counter32;
41     config false;
42     description
43         "Counter that increments every time a Pdelay_Resp message
44         is transmitted.";
45     reference
46         "14.12.17 of IEEE Std 802.1AS";
47 }
48 leaf tx-pdelay-resp-follow-up-count {
49     type yang:counter32;
50     config false;
51     description
52         "Counter that increments every time a
53         Pdelay_Resp_Follow_Up message is transmitted.";
54     reference
55         "14.12.18 of IEEE Std 802.1AS";
56 }
57 leaf tx-announce-count {
58     type yang:counter32;
59     config false;

```

```

1      description
2      "Counter that increments every time an Announce message is
3      transmitted.";
4      reference
5      "14.12.19 of IEEE Std 802.1AS";
6  }
7 }
8 }
9
10 augment
11     "/ptp-tt:ptp"+
12     "/ptp-tt:instances"+
13     "/ptp-tt:instance"+
14     "/ptp-tt:ports"+
15     "/ptp-tt:port" {
16     description
17     "Augment to add asymmetry-measurement-mode-ds to IEEE Std 1588
18     PTP Port.";
19     container asymmetry-measurement-mode-ds {
20     description
21     "Represents the capability to enable/disable the Asymmetry
22     Compensation Measurement Procedure on a PTP Port. This data
23     set is used instead of the CMLDS
24     asymmetry-measurement-mode-ds when only a single PTP
25     Instance is present (i.e., CMLDS is not used).";
26     reference
27     "14.15 of IEEE Std 802.1AS
28     Annex G of IEEE Std 802.1AS";
29     leaf enabled {
30     type boolean;
31     description
32     "For full-duplex IEEE Std 802.3 media, the value is true
33     if an asymmetry measurement is being performed for the
34     link attached to this PTP Port, and false otherwise. For
35     all other media, the value shall be false.";
36     reference
37     "14.15.2 of IEEE Std 802.1AS";
38     }
39     }
40 }
41
42 augment
43     "/ptp-tt:ptp"+
44     "/ptp-tt:common-services"+
45     "/ptp-tt:cmlDs"+
46     "/ptp-tt:ports"+
47     "/ptp-tt:port"+
48     "/ptp-tt:link-port-ds" {
49     description
50     "Augment IEEE Std 1588 cmlDsLinkPortDS.";
51     leaf cmlDs-link-port-enabled {
52     type boolean;
53     config false;
54     description
55     "Boolean that is true if both delay-mechanism is common-p2p
56     and the value of ptp-port-enabled is true, for at least one
57     PTP Port that uses the CMLDS; otherwise, the value is
58     false.";
59     reference

```



```

1      "11.2.18.1 of IEEE Std 802.1AS
2      14.18.3 of IEEE Std 802.1AS";
3  }
4  leaf is-measuring-delay {
5      type boolean;
6      config false;
7      description
8          "This leaf is analogous to is-measuring-delay for a PTP
9          Port, but applicable to this Link Port.";
10     reference
11         "14.18.4 of IEEE Std 802.1AS";
12 }
13 leaf as-capable-across-domains {
14     type boolean;
15     config false;
16     description
17         "This leaf is true when all PTP Instances (domains) for this
18         Link Port detect proper exchange of Pdelay messages.";
19     reference
20         "11.2.2 of IEEE Std 802.1AS
21         14.18.5 of IEEE Std 802.1AS";
22 }
23 leaf mean-link-delay-thresh {
24     type ptp-tt:time-interval;
25     description
26         "Propagation time threshold for mean-link-delay, above which
27         a Link Port is not considered capable of participating in
28         the IEEE Std 802.1AS protocol.";
29     reference
30         "14.18.7 of IEEE Std 802.1AS";
31 }
32 leaf neg-mean-link-delay-thresh {
33     type ptp-tt:time-interval;
34     description
35         "The negative propagation time threshold for
36         mean-link-delay, below which a Link Port is not considered
37         capable of participating in the IEEE Std 802.1AS
38         protocol.";
39     reference
40         "14.18.8 of IEEE Std 802.1AS";
41 }
42 leaf initial-log-pdelay-req-interval {
43     type int8;
44     description
45         "This leaf is analogous to initial-log-pdelay-req-interval
46         for a PTP Port, but applicable to this Link Port.";
47     reference
48         "14.18.11 of IEEE Std 802.1AS";
49 }
50 leaf current-log-pdelay-req-interval {
51     type int8;
52     config false;
53     description
54         "This leaf is analogous to current-log-pdelay-req-interval
55         for a PTP Port, but applicable to this Link Port.";
56     reference
57         "14.18.12 of IEEE Std 802.1AS";
58 }
59 leaf use-mgt-log-pdelay-req-interval {

```

```

1      type boolean;
2      description
3          "This leaf is analogous to use-mgt-log-pdelay-req-interval
4          for a PTP Port, but applicable to this Link Port.";
5      reference
6          "14.18.13 of IEEE Std 802.1AS";
7  }
8  leaf mgt-log-pdelay-req-interval {
9      type int8;
10     description
11         "This leaf is analogous to mgt-log-pdelay-req-interval for
12         a PTP Port, but applicable to this Link Port.";
13     reference
14         "14.18.14 of IEEE Std 802.1AS";
15 }
16 leaf initial-compute-neighbor-rate-ratio {
17     type boolean;
18     description
19         "This leaf is analogous to
20         initial-compute-neighbor-rate-ratio for a PTP Port, but
21         applicable to this Link Port.";
22     reference
23         "14.18.15 of IEEE Std 802.1AS";
24 }
25 leaf current-compute-neighbor-rate-ratio {
26     type boolean;
27     config false;
28     description
29         "This leaf is analogous to
30         current-compute-neighbor-rate-ratio for a PTP Port, but
31         applicable to this Link Port.";
32     reference
33         "14.18.16 of IEEE Std 802.1AS";
34 }
35 leaf use-mgt-compute-neighbor-rate-ratio {
36     type boolean;
37     description
38         "This leaf is analogous to
39         use-mgt-compute-neighbor-rate-ratio for a PTP Port, but
40         applicable to this Link Port.";
41     reference
42         "14.18.17 of IEEE Std 802.1AS";
43 }
44 leaf mgt-compute-neighbor-rate-ratio {
45     type boolean;
46     description
47         "This leaf is analogous to mgt-compute-neighbor-rate-ratio
48         for a PTP Port, but applicable to this Link Port.";
49     reference
50         "14.18.18 of IEEE Std 802.1AS";
51 }
52 leaf initial-compute-mean-link-delay {
53     type boolean;
54     description
55         "This leaf is analogous to initial-compute-mean-link-delay
56         for a PTP Port, but applicable to this Link Port.";
57     reference
58         "14.18.19 of IEEE Std 802.1AS";
59 }

```

```

1  leaf current-compute-mean-link-delay {
2      type boolean;
3      config false;
4      description
5          "This leaf is analogous to current-compute-mean-link-delay
6          for a PTP Port, but applicable to this Link Port.";
7      reference
8          "14.18.20 of IEEE Std 802.1AS";
9  }
10 leaf use-mgt-compute-mean-link-delay {
11     type boolean;
12     description
13         "This leaf is analogous to use-mgt-compute-mean-link-delay
14         for a PTP Port, but applicable to this Link Port.";
15     reference
16         "14.18.21 of IEEE Std 802.1AS";
17 }
18 leaf mgt-compute-mean-link-delay {
19     type boolean;
20     description
21         "This leaf is analogous to mgt-compute-mean-link-delay for a
22         PTP Port, but applicable to this Link Port.";
23     reference
24         "14.18.22 of IEEE Std 802.1AS";
25 }
26 leaf allowed-lost-responses {
27     type uint8;
28     description
29         "This leaf is analogous to allowed-lost-responses for a PTP
30         Port, but applicable to this Link Port.";
31     reference
32         "14.18.23 of IEEE Std 802.1AS";
33 }
34 leaf allowed-faults {
35     type uint8;
36     description
37         "This leaf is analogous to allowed-faults for a PTP Port,
38         but applicable to this Link Port.";
39     reference
40         "14.18.24 of IEEE Std 802.1AS";
41 }
42 leaf-list pdelay-truncated-timestamps {
43     type uinteger48;
44     config false;
45     description
46         "This leaf is analogous to pdelay-truncated-timestamps for a
47         PTP Port, but applicable to this Link Port.";
48     reference
49         "14.18.26 of IEEE Std 802.1AS";
50 }
51 }
52
53 augment
54     "/ptp-tt:ptp"+
55     "/ptp-tt:common-services"+
56     "/ptp-tt:cmls"+
57     "/ptp-tt:ports"+
58     "/ptp-tt:port" {
59     description

```

```
1      "Augment to add port-statistics-ds to IEEE Std 1588 Link
2      Port.";
3      container port-statistics-ds {
4          description
5              "This container is analogous to port-statistics-ds for a PTP
6              Port, but applicable to this Link Port.";
7          reference
8              "14.19 of IEEE Std 802.1AS";
9          leaf rx-pdelay-req-count {
10             type yang:counter32;
11             config false;
12             description
13                 "This leaf is analogous to rx-pdelay-req-count for a PTP
14                 Port, but applicable to this Link Port.";
15             reference
16                 "14.19.2 of IEEE Std 802.1AS";
17         }
18         leaf rx-pdelay-resp-count {
19             type yang:counter32;
20             config false;
21             description
22                 "This leaf is analogous to rx-pdelay-resp-count for a PTP
23                 Port, but applicable to this Link Port.";
24             reference
25                 "14.19.3 of IEEE Std 802.1AS";
26         }
27         leaf rx-pdelay-resp-follow-up-count {
28             type yang:counter32;
29             config false;
30             description
31                 "This leaf is analogous to rx-pdelay-resp-follow-up-count
32                 for a PTP Port, but applicable to this Link Port.";
33             reference
34                 "14.19.4 of IEEE Std 802.1AS";
35         }
36         leaf rx-packet-discard-count {
37             type yang:counter32;
38             config false;
39             description
40                 "This leaf is analogous to rx-packet-discard-count for a
41                 PTP Port, but applicable to this Link Port.";
42             reference
43                 "14.19.5 of IEEE Std 802.1AS";
44         }
45         leaf pdelay-allowed-lost-exceeded-count {
46             type yang:counter32;
47             config false;
48             description
49                 "This leaf is analogous to
50                 pdelay-allowed-lost-exceeded-count for a PTP Port, but
51                 applicable to this Link Port.";
52             reference
53                 "14.19.6 of IEEE Std 802.1AS";
54         }
55         leaf tx-pdelay-req-count {
56             type yang:counter32;
57             config false;
58             description
59                 "This leaf is analogous to tx-pdelay-req-count for a
```

```

1         PTP Port, but applicable to this Link Port.";
2     reference
3         "14.19.7 of IEEE Std 802.1AS";
4 }
5 leaf tx-pdelay-resp-count {
6     type yang:counter32;
7     config false;
8     description
9         "This leaf is analogous to tx-pdelay-resp-count for a
10        PTP Port, but applicable to this Link Port.";
11    reference
12        "14.19.8 of IEEE Std 802.1AS";
13 }
14 leaf tx-pdelay-resp-follow-up-count {
15     type yang:counter32;
16     config false;
17     description
18         "This leaf is analogous to tx-pdelay-resp-follow-up-count
19        for a PTP Port, but applicable to this Link Port.";
20    reference
21        "14.19.9 of IEEE Std 802.1AS";
22 }
23 }
24 }
25
26 augment
27     "/ptp-tt:ptp"+
28     "/ptp-tt:common-services"+
29     "/ptp-tt:cmls"+
30     "/ptp-tt:ports"+
31     "/ptp-tt:port" {
32     description
33         "Augment to add asymmetry-measurement-mode-ds to IEEE
34        Std 1588 Link Port.";
35     container asymmetry-measurement-mode-ds {
36         description
37             "This container is analogous to
38            asymmetry-measurement-mode-ds for a PTP Port, but
39            applicable to this Link Port.";
40         reference
41             "14.20 of IEEE Std 802.1AS";
42         leaf enabled {
43             type boolean;
44             description
45                 "This leaf is analogous to
46                asymmetry-measurement-mode-ds.enabled for a PTP Port, but
47                applicable to this Link Port.";
48             reference
49                 "14.20.2 of IEEE Std 802.1AS";
50         }
51     }
52 }
53 }
54
55

```

1 **Insert the following new Clause 19:**

2 **19. Media-dependent layer specification for IEEE Std 802.3 Clause 4 Media** 3 **Access Control (MAC) operating in half-duplex**

4 **19.1 Overview**

5 **19.1.1 General**

6 This clause specifies the media-dependent layer that provides synchronized time across links using the IEEE
 7 Std 802.3 Clause 4 MAC operating in half-duplex mode

8 **19.1.1.1 Half-duplex Ethernet (HDE) characteristics**

9 The IEEE Std 802.3 Clause 4 MAC can operate in either full-duplex or half-duplex mode. When this MAC
 10 is operating in full-duplex, its media-dependent specification for gPTP is covered in Clause 11. Clause 19 is
 11 used when the IEEE Std 802.3 Clause 4 MAC is operating in half-duplex. This mode necessitates additional
 12 managed object settings and frame processing due to the effects of the shared media this mode supports,
 13 which are described in this clause.

14 Figure 19-1 shows an example of time-aware network using HDE links.

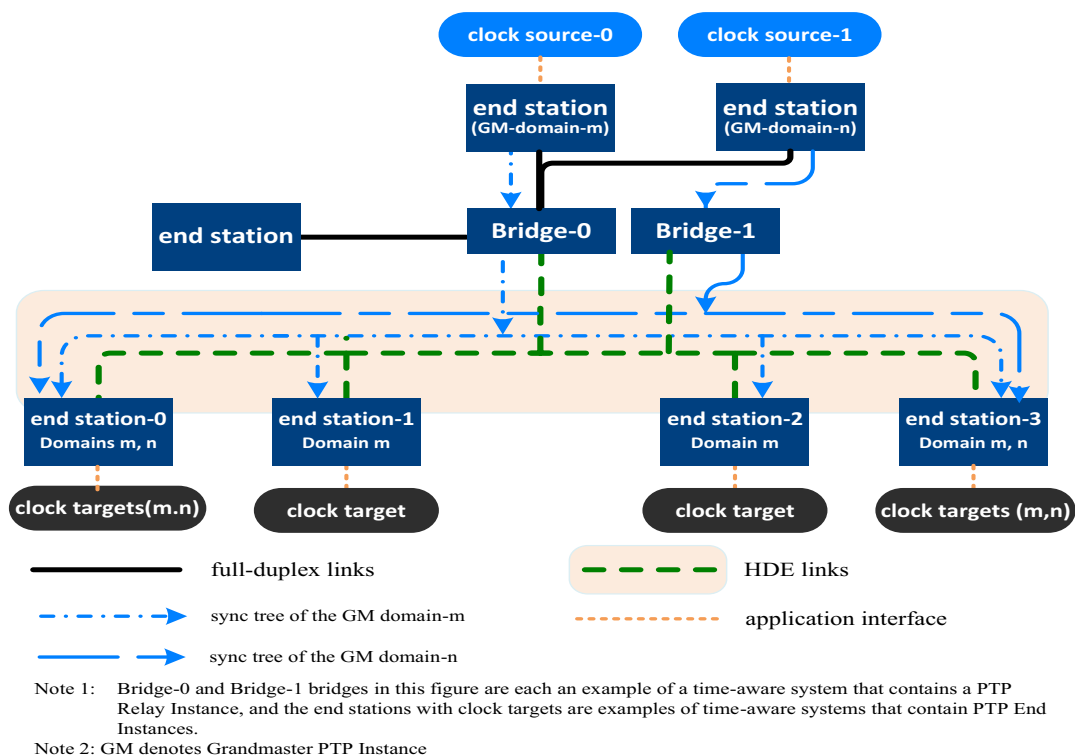


Figure 19-1— Time-aware network example where the four end stations near the bottom of the figure are connected to the bridge via an HDE link

1 When using HDE links, the use of “neighbor” refers to a device or to devices (end station or bridge) that is/2 are the intended recipient(s) of a transmitted PTP message. The intended recipient(s) of a PTP message from3 a gPTP TimeTransmitter port is/are one or more gPTP TimeReceiver port(s). The intended recipient of a4 PTP message from a gPTP TimeReceiver port is the gPTP TimeTransmitter port, regardless of whether or5 not other gPTP TimeTransmitter ports received that message (e.g., if all ports are connected via a shared6 medium). In the example in Figure 19-1, PTP messages are exchanged between a gPTP TimeTransmitter7 port of a bridge (e.g., Bridge-0) and the four gPTP TimeReceiver ports of the four end stations via a shared8 medium.

9 NOTE—For example, this standard supports 10BASE-T1S, as specified in Clause 147 of IEEE Std 802.3, utilizing10 Physical Layer Collision Avoidance (PLCA), as specified in Clause 148 of IEEE Std 802.3.

11 19.1.1.2 Overview of the major differences and restrictions of using HDE links

12 This standard uses Clause 11 in its entirety with the following differences (complete details follow starting13 in 19.1.2):

- 14 a) The peer delay initiator is restricted to timeReceivers only (see 19.1.2).
- 15 b) External port configuration mode is the only mode supported (see 19.8).

16 The following features of this standard have not been qualified for use with HDE links:

- 17 c) One-step time transport (see 19.1.3 and 19.2.16)
- 18 d) CMLDS (see 19.2.17)
- 19 e) The use of Signaling messages (see 19.8)
- 20 f) Hot Standby (see 19.8)

21 19.1.2 Propagation delay measurement over HDE links

22 The measurement of propagation delay on an HDE link using the peer-to-peer delay mechanism is23 illustrated in Figure 11-1 and is described in 11.1.2, with the following differences:

- 24 a) The peer delay initiator is restricted to each timeReceiver port
- 25 b) The timeTransmitter port does not initiate the peer-to-peer delay mechanism.

26 Therefore, pDelayReqSendDisabled and pDelayRespSendDisabled are set as follows:

- 27 c) pDelayReqSendDisabled is set to FALSE when the PTP Port state is TimeReceiverPort and is28 otherwise set to TRUE.
- 29 d) pDelayRespSendDisabled is set to FALSE when the PTP Port state is TimeTransmitterPort and is30 otherwise set to TRUE

31 Other configuration values for item c) and item d) are not intended for use with HDE links by this standard.

32 19.1.3 Transport of time-synchronization information

33 HDE links uses two-step time transport as specified in clause 11.1.3. The variables oneStepReceive (see34 19.2.13.9), oneStepTransmit (see 19.2.13.10), and oneStepTxOper(see 19.2.13.11) are set to FALSE for35 HDE links. Other configuration values for oneStepTransmit and oneStepTxOper are not intended for use36 with HDE links by this standard.

37 NOTE—The transport of time-synchronization information by a PTP Instance, using Sync and Follow_Up messages, is38 illustrated in Figure 11-2.

1 19.1.4 Model of operation

2 The model for a PTP Instance of a time-aware system with full-duplex point-to-point links is shown in
3 Figure 11-3. This clause reuses Figure 11-3 (as its structure is unchanged for this clause), where all
4 references to clause 11 in Figure 11-3 are to be replaced by references to clause 19 (this clause). The
5 presence of one HDE MD entity per PTP Port is assumed.

6 A general, media-independent description of the generation of timestamps is given in 8.4.3. A more specific
7 description for PTP event messages is given in 11.3.2.1. A PTP event message is timestamped relative to the
8 LocalClock entity when the message timestamp point (see 3.17) crosses the timestamp measurement plane
9 (see 3.33). The timestamp is corrected for any ingressLatency or egressLatency (see 8.4.3) to produce a
10 timestamp relative to the reference plane (see 3.26). The corrected timestamp value is provided to the MD
11 entity.

12 The MD entity behavior and detailed state machines specific to full-duplex point-to-point links, which are
13 described in 11.2, are reused for HDE links subject to the conditions defined in 19.2.

14 19.2 State machines for HDE links**15 19.2.1 General**

16 The state machines for HDE links are described in 11.2.1 except that there is one instance of the
17 MDPdelayReq state machine per PTP Port, there is one instance of the MDPdelayResp state machine per
18 port, and the LinkDelayIntervalSetting state machine is not used.

19 19.2.2 Determination of asCapable and asCapableAcrossDomains

20 As specified in 11.2.2.

21 NOTE—For full-duplex point-to-point links, asCapableAcrossDomains is one global variable for all domains per link
22 port. For HDE links, asCapableAcrossDomains is a global variable per PTP Instance (i.e., per domain), the name
23 asCapableAcrossDomains has been kept for backwards compatibility with existing implementations.

24 19.2.3 Use of MAC Control PAUSE operation

25 As specified in 11.2.3.

26 19.2.4 Use of priority-based flow control

27 As specified in 11.2.4.

28 19.2.5 Use of link aggregation

29 As specified in 11.2.5.

**30 19.2.6 Service interface primitives and data structures communicated between state
31 machines**

32 The following two subclauses describe the service primitives and data structures communicated between the
33 time-synchronization state machines of the MD entity. First the service primitives are described, followed by
34 the data structures.

1 **19.2.7 DL-UNITDATA.request**

2 This service primitive is as specified in 11.2.7.

3 **19.2.8 DL-UNITDATA.indication**

4 This service primitive is as specified in 11.2.8.

5 **19.2.9 MDTimestampReceive**

6 This data structure is as specified in 11.2.9.

7 **19.2.10 MDSyncReceive**

8 This data structure is as specified in 10.2.2.2.

9 **19.2.11 MDSyncSend**

10 This structure is as specified in 10.2.2.1.

11 **19.2.12 Overview of MD entity global variables**

12 The overview of MD entity global variables is given in 11.2.12.

13 **19.2.13 MD entity global variables**

14 **19.2.13.1 currentLogPdelayReqInterval:** This variable is as specified in 11.2.13.1. However, for HDE,
15 there is one instance of this variable per PTP Port.

16 **19.2.13.2 initialLogPdelayReqInterval:** This variable is as specified in 11.2.13.2. However, for HDE, there
17 is one instance of this variable per PTP Port.

18 **19.2.13.3 pdelayReqInterval:** This variable is as specified in 11.2.13.3. However, for HDE, there is one
19 instance of this variable per PTP Port.

20 **19.2.13.4 allowedLostResponses:** This variable is as specified in 11.2.13.4. However, for HDE, there is one
21 instance of this variable per PTP Port.

22 **19.2.13.5 allowedFaults:** This variable is as specified in 11.2.13.5. However, for HDE, there is one instance
23 of this variable per PTP Port.

24 **19.2.13.6 isMeasuringDelay:** This variable is as specified in 11.2.13.6. However, for HDE, the port
25 measures propagation delay if it is receiving Pdelay_Resp and Pdelay_Resp_Follow_Up messages from the
26 requested port with the requested clockIdentity. However, for HDE, there is one instance of this variable per
27 PTP Port.

28 **19.2.13.7 meanLinkDelayThresh:** This variable is as specified in 11.2.13.7. However, for HDE, there is
29 one instance of this variable per PTP Port.

30 **19.2.13.8 negMeanLinkDelayThresh:** This variable is as specified in 11.2.13.8. However, for HDE, there
31 is one instance of this variable per PTP Port.

32 **19.2.13.9 syncSequenceId:** This variable is as specified in 11.2.13.8.

1 **19.2.13.10 oneStepReceive**: This variable is as specified in 11.2.13.9. However, for HDE, this variable is
2 FALSE.

3 **19.2.13.11 oneStepTransmit**: This variable is as specified in 11.2.13.10. However, for HDE, this variable is
4 FALSE.

5 **19.2.13.12 oneStepTxOper**: This variable is as specified in 11.2.13.11. However, for HDE, this variable is
6 FALSE.

7 **19.2.13.13 asCapableAcrossDomains**: This variable is as specified in 11.2.13.12.

8 **19.2.13.14 nrrPdelay**: This variable is as specified in 11.2.13.13.

9 **19.2.13.15 nrrSync**: This variable is as specified in 11.2.13.14.

10 **19.2.13.16 nrrCompMethod**: This variable is as specified in 11.2.13.15.

11 **19.2.14 MDSyncReceiveSM state machine**

12 The MDSyncReceiveSM state machine is as specified in 11.2.14.

13 **19.2.15 MDSyncSendSM state machine**

14 The MDSyncSendSM state machine is as specified in 11.2.15.

15 **19.2.16 OneStepTxOperSetting state machine**

16 This state machine is not used for HDE links.

17 **19.2.17 Common Mean Link Delay Service (CMLDS)**

18 CMLDS has not been qualified for use with HDE links.

19 HDE links use the transport-specific peer-to-peer delay mechanism for each domain. Therefore, if the time-
20 aware system implements multiple domains on an HDE link, each domain uses the transport-specific peer-
21 to-peer delay mechanism on that link regardless of its domain number.

22 If multiple TimeTransmitter ports are present on an HDE link, they are in different gPTP domains. CMLDS
23 cannot be used because, in general, the TimeTransmitter ports can be on physical ports of different time-
24 aware systems (i.e., different bridges). In this case, both meanLinkDelay and neighborRateRatio between a
25 PTP End Instance in one of the domains and the TimeTransmitter it is communicating with can be different
26 from meanLinkDelay and neighborRateRatio between a PTP End Instance in another domain on the same
27 end station and the TimeTransmitter that PTP End Instance is communicating with. For instance, in Figure
28 19-1 the meanLinkDelay and neighborRateRatio between the PTP End Instance over domain-m and over
29 domain-n at end station-0 are computed using timestamps from two distinct time aware systems, GM-
30 domain-m and GM-domain-n, respectively; and hence does not use CMLDS.

31 **19.2.18 Common Mean Link Delay Service (CMLDS) global variable**

32 The Common Mean Link Delay Service (CMLDS) global variable is as specified in 11.2.18.

33 **19.2.19 MDPdelayReq state machine**

34 The MDPdelayReq state machine is as specified in 11.2.19.

1 The variable `pdelayReqSendDisabled` is set per 19.1.2.

2 `Pdelay_Req`, `Pdelay_Resp`, and `Pdelay_Resp_Follow_Up` messages shall be transmitted with
3 `domainNumber` of the HDE gPTP domain used by the transport-specific peer delay mechanism (see 19.4).
4 Any received `Pdelay_Req`, `Pdelay_Resp`, and `Pdelay_Resp_Follow_Up` messages are ignored by the state
5 machines if the `domainNumber` in the received message does not match the gPTP domain associated with
6 the PTP Port.

7 **19.2.20 MDPdelayResp state machine**

8 The MDPdelayResp state machine is as specified in 11.2.20.

9 The variable `pdelayRespSendDisabled` is set per 19.1.2.

10 `Pdelay_Req`, `Pdelay_Resp`, and `Pdelay_Resp_Follow_Up` messages shall be transmitted with
11 `domainNumber` of the HDE gPTP domain used by the transport-specific peer delay mechanism (see 19.4).
12 Any received `Pdelay_Req`, `Pdelay_Resp`, and `Pdelay_Resp_Follow_Up` messages are ignored by the state
13 machines if the `domainNumber` in the received message does not match the gPTP domain associated with
14 the PTP Port.

15 **19.2.21 LinkDelayIntervalSetting state machine**

16 This state machine is not used for HDE links.

17 **19.3 Message attributes**

18 Message attributes is as specified in 11.3.

19 **19.4 Message formats**

20 Message formats is as specified in 11.4, except for 11.4.2.4, where the `Pdelay_Req`, `Pdelay_Resp`, and
21 `Pdelay_Resp_Follow_Up` messages shall be transmitted with the `domainNumber` of the HDE gPTP domain
22 used by the transport-specific peer delay mechanism. Any `Pdelay_Req`, `Pdelay_Resp`, and
23 `Pdelay_Resp_Follow_Up` message received with a `domainNumber` other than the `domainNumber` of the
24 HDE gPTP domain used by the transport-specific peer delay mechanism shall be ignored. The
25 `domainNumber` for all other PTP messages is as specified in 10.6.2.2.6.

26 **19.5 Protocol timing characterization**

27 **19.5.1 General**

28 This subclause specifies timing attributes for the media-dependent sublayer specified in this clause.

29 **19.5.2 Message transmission intervals**

30 **19.5.2.1 General interval specification**

31 The general message transmission interval specification is as specified in 11.5.2.1.

1 19.5.2.2 Pdelay_Req message transmission interval

2 Pdelay_Req message transmission interval is as specified in 11.5.2.2. The variable
3 useMgtSettableLogPdelayReqInterval is set to TRUE. Other configuration values for
4 useMgtSettableLogPdelayReqInterval are not intended for use with HDE links by this standard.

5 19.5.2.3 Sync message transmission interval default value

6 The Sync message transmission interval default value is as specified in 11.5.2.3.

7 19.5.3 allowedLostResponses

8 The variable allowedLostResponses is as specified in 11.5.3.

9 19.5.4 allowedFaults

10 The variable allowedFaults is as specified in 11.5.4.

11 19.6 Control of computation of neighborRateRatio

12 The control of computation of neighborRateRatio is as specified in 11.6.

13 19.7 Control of computation of meanLinkDelay

14 The control of computation of meanLinkDelay is as specified in 11.7.

15 19.8 HDE links settings and configuration

16 This subclause provides settings and configurations that are specific for HDE links, other configuration
17 values are not intended for use with HDE links by this standard.

- 18 a) The per PTP Instance global variable externalPortConfigurationEnabled is set to TRUE for HDE
19 links.
- 20 b) Both GtpCapableTransmit and GtpCapableReceive state machines are disabled for HDE links,
21 and therefore gtpCapableStateMachinesEnabled are set to FALSE (see 10.4.1). According to
22 10.4.1, if the managed object gtpCapableStateMachinesEnabled is FALSE, the global variable
23 neighborGtpCapable for the port (see 10.2.5.16) is set to TRUE.
- 24 c) The variables useMgtSettableLogSyncInterval and useMgtSettableLogAnnounceInterval are set to
25 TRUE. As a consequence, the SyncIntervalSetting and the AnnounceIntervalSetting state machines
26 are not used for HDE links.
- 27 d) The GtpCapableIntervalSetting state machine is not used for HDE links. The variable
28 useMgtSettableLogGtpCapableMessageInterval is set to TRUE.

29 Hot Standby (see Clause 18) has not been qualified for use with HDE links.

Annex A

(normative)

Protocol Implementation Conformance Statement (PICS) proforma³

Add a row at the end of Table A.5 as follows:

A.5 Major Capabilities

Item	Feature	Status	References	Support
MDHDE	Does the PTP Instance support media-dependent HDE link functionality on one or more PTP Ports?	O:1	5.9, 11, 19, A.6, A.22	Yes [] No []

Change A.6 as follows:

A.6 Media access control methods

Item	Feature	Status	References	Support
MAC-IEEE-802.3	Which MAC methods are implemented in conformance with the relevant MAC standards?	O:2	11.1 19.1	Yes [] No [] Yes [] No []
MAC-IEEE-802.11		O:2	12.1	Yes [] No []
MAC-1	Has a PICS been completed for each of the MAC methods implemented as required by the relevant MAC Standards?	M		Yes []
MAC-2	Do all the MAC methods implemented support the MAC Timing aware Service as specified?	M	Clause 11 Clause 12 Clause 13 Clause 19	Yes []

³ Copyright release for PICS proformas: Users of this standard may freely reproduce the PICS proforma in this annex so that it can be used for its intended purpose and may further publish the completed PICS.

Change A.21 as follows:**A.21 External port configuration**

Item	Feature	Status	References	Support
	If item EXT is not supported, mark N/A.			NA []
EXT-1	Does the PTP Instance support the specifications for externalPortConfigurationEnabled value of true?	EXT:M	10.3.1 19.8	Yes []
EXT-2	Does the PTP Instance support the PortAnnounceInformationExt state machine?	EXT:M	10.3.14	Yes []
EXT-3	Does the PTP Instance support the PortStateSettingExt state machine?	EXT:M	10.3.15	Yes []

Insert Table A.22 as follows:**A.22 Media-dependent, HDE link**

Item	Feature	Status	References	Support
MDHDE-1	Does this PTP Port implement the functionality of the MDSyncReceiveSM state machine in compliance with the requirements of 19.2.14, 11.2.14 and Figure 11-6?	MDHDE:M	19.2.14, 11.2.14	Yes []
MDHDE-2	Does this PTP Port implement the functionality of the MDSyncSendSM state machine in compliance with the requirements of 19.2.15, 11.2.15 and Figure 11-7?	MIMSTR and MDHDE:M	19.2.15, 11.2.15	Yes []
MDHDE-3	Does this port implement the functionality of the MDPdelayRequest state machine in compliance with the requirements of 19.1.2, 19.2.19, 11.2.19 and Figure 11-9?	MDHDE:M	19.1.2, 19.2.19, 11.2.19	Yes []
MDHDE-4	Does this port implement the functionality of the MDPdelayResponse state machine in compliance with the requirements of 19.1.2, 19.2.20, 11.2.20 and Figure 11-10?	MDHDE:M,	19.2.20 19.1.2 11.2.20	Yes []
MDHDE-5	Does this PTP Port timestamp Sync messages on ingress with respect to the LocalClock in compliance with 19.3, 11.3.2.1 and 11.3.9?	MDHDE:M	19.3, 11.3.2.1	Yes []
MDHDE-6	Does this PTP Port timestamp Sync messages on egress with respect to the LocalClock in compliance with the requirements of 19.3, 11.3.2.1 and 11.3.9?	MIMSTR and MDHDE:M	19.3 11.3.2.1	Yes []
MDHDE-7	Does this port timestamp Pdelay_Req messages on ingress and egress with respect to the LocalClock in compliance with the requirements of 19.3, 11.3.2.1 and 11.3.9?	MDHDE:M	19.3, 11.3.2.1	Yes []
MDHDE-8	Does this port timestamp Pdelay_Resp messages on ingress and egress with respect to the LocalClock in compliance with the requirements of 19.3, 11.3.2.1 and 11.3.9?	MDHDE:M	19.3, 11.3.2.1	Yes []

A.22 Media-dependent, HDE link (continued)

Item	Feature	Status	References	Support
MDHDE-9	Are all IEEE 802.1AS messages on this port sent without a Q-tag in compliance with the requirements of 19.3, 11.3.3?	MDHDE:M	19.3, 11.3.3	Yes []
MDHDE-10	Do all media-dependent messages transmitted on this port use a destination MAC address taken from Table 11-3 in compliance with the requirements of 19.3 and 11.3.4 [01-80-C2-00-00-0E]?	MDHDE:M	19.3, 11.3.4	Yes []
MDHDE-11	Do all media-dependent messages transmitted on this port use a source MAC address that is assigned to that port in compliance with the requirements of 19.3 and 11.3.4?	MDHDE:M	19.3, 11.3.4	Yes []
MDHDE-12	Do all media-dependent message transmitted on this port use an EtherType specified in Table 11-4 [88-F7]?	MDHDE:M	19.3, 11.3.5	Yes []
MDHDE-13	Does the header of all the media-dependent messages on this port comply with the requirements of 19.4, 11.4.2 and Table 10-7?	MDHDE:M	19.4, 11.4.2	Yes [] N/A []
MDHDE-14	Does the body of Sync messages sent on this PTP Port comply with the requirements of 19.4, 11.4.3, Table 11-8, and Table 11-9?	MDHDE:M	19.4, 11.4.3	Yes []
MDHDE-15	Does the body of Follow_Up messages sent on this PTP Port comply with the requirements of 19.4, 11.4.4, 6.4.3.3 (lastGmPhaseChange), and Table 11-10?	MDHDE:M	19.4, 11.4.4, 6.4.3.3	Yes []
MDHDE-16	Does the body of Pdelay_Req messages sent on this port comply with the requirements of 19.4, 11.4.5 and Table 11-12?	MDHDE:M	19.4, 11.4.5	Yes []
MDHDE-17	Does the body of Pdelay_Resp messages sent on this port comply with the requirements of 19.4, 11.4.6 and Table 11-13?	MDHDE:M	19.4, 11.4.6	Yes []
MDHDE-18	Does the body of Pdelay_Resp_Follow_Up messages sent on this port comply with the requirements of 19.4, 11.4.7 and Table 11-14?	MDHDE:M	19.4, 11.4.7	Yes []
MDHDE-19	Are all reserved fields in media-dependent messages sent on this port set to 0 in compliance with the requirements of 19.4, 11.4.1?	MDHDE:M	19.4, 11.4.1	Yes []
MDHDE-20	Do the Sync message sequence numbers comply with the requirements of 19.3, 11.3.8?	MIMSTR and MDHDE:M	19.3, 11.3.8	Yes [] N/A []
MDHDE-21	Do the Pdelay_Req message sequence numbers comply with the requirements of 19.3 and 11.3.8?	MDHDE:M	19.3, 11.3.8	Yes []

A.22 Media-dependent, HDE link (continued)

Item	Feature	Status	References	Support
MDHDE-22	Does the Pdelay mean request transmission interval comply with the requirements of 19.5.2.2 and 11.5.2.2?	MDHDE:M	19.5.2.2, 11.5.2.2	Yes []
MDHDE-23	Does the Sync mean transmission interval comply with the requirements of 19.5.2.3 and 11.5.2.3?	MDHDE:M	19.5.2.3, 11.5.2.3	Yes []
MDHDE-24	Does the HDE media-dependent layer set the asCapable global variable in the media-independent PortSync entity in compliance with the requirements of 19.2.2 and 11.2.2?	MDHDE:M	19.2.2, 11.2.2	Yes []
MDHDE-25	Does the PTP Instance consider the PTP Port or Link Port, respectively, to not be exchanging Pdelay messages when a valid response is not received in compliance with the requirements of 19.5.3 and 11.5.3?	MDHDE:M	19.5.3, 11.5.3	Yes []
MDHDE-26	Does the PTP Instance ignore TLVs, of PTP messages, that it cannot parse and attempt to parse the next TLV, in compliance with the requirements of 19.4 and 11.4.1?	MDHDE:M	19.4, 11.4.1	Yes []
MDHDE-27	Does the time-aware system initialize meanLinkDelayThresh as specified in 19.2.2 and 11.2.2?	MDHDE:M	19.2.2, 11.2.2	Yes []
MDHDE-28	Does this port support propagation delay averaging?	MDHDE:O	19.2.19, 11.2.19.3.4	Yes [] No []
MDHDE-29	Does this port support two-step capability on receive?	MDHDE:M	19.2.14, 11.2.14, item d) of 5.5	Yes []
MDHDE-30	Does this port support two-step capability on transmit?	MDHDE:M	19.2.15, 11.2.15, item e) of 5.5	Yes []
MDHDE-31	Is the transport-specific peer-to-peer delay mechanism supported?	MDHDE:M	19.2.17	Yes []

¹ **Annex F**

² (informative)

³ **PTP profile included in this standard**

⁴ **F.1 General**

⁵ *Change the paragraph in F.1 as follows:*

⁶ The specification in this standard of synchronized time transport over a full-duplex point-to-point link
⁷ includes a PTP profile. The information contained in a PTP profile is described in 20.3 of IEEE Std 1588-
⁸ 2019. This annex summarizes the PTP profile for transport of timing over full-duplex point-to-point links.
⁹ This PTP profile is also used in the transport of timing over HDE links (see Clause 19), and in the transport
¹⁰ of timing over CSN when a CSN clock reference is not present (see Clause 16). This PTP profile is not used
¹¹ in the transport of timing over IEEE Std 802.11 links and IEEE Std 802.3 EPON links; both these transports
¹² use native timing mechanisms to assist in the synchronized time transport.

¹³